

An integrative framework for the human sense of control

Alireza Modirshanechi^{1,2,*}, Peter Dayan^{2,3,†}, and Eric Schulz^{1,†}

¹ Helmholtz Munich, Munich, Germany

² Max Planck Institute for Biological Cybernetics, Tübingen, Germany

³ University of Tübingen, Tübingen, Germany

* Corresponding author: alireza.modirshanechi@helmholtz-munich.de

† These authors contributed equally to this work

Abstract

From managing daily schedules to making major life decisions, how much we feel in control of our lives is central to our behavior and well-being. Yet, there is currently no precise, unifying account of what it means to feel in control. Here, we propose that humans feel in control when they are prepared to tackle future tasks. We formalize this perspective via a precise yet interpretable measure of controllability that quantifies three key factors: (i) the range of available actions, (ii) the certainty or (iii) the possibility of achieving desired outcomes. This measure unifies previous conceptions of control and successfully predicts human preferences in three experiments in which participants either rate their sense of control or prepare themselves to seek rewards or avoid punishments. Our framework offers a principled foundation for understanding how environmental structure shapes the human sense of control, paving the way towards an integrative science of agency.

Introduction

In everyday life, we constantly navigate uncertain environments, attempting to steer the course of events in our favor. Consider a familiar situation: planning your weekend. You may have several activities in mind, from attending a painting class to going on a hike. At times, all your options may be accessible; at others, only a subset may be available, or availability may shift unexpectedly. Even when the number of available options remains the same, the structure of the environment, its predictability, and the extent to which your choices seem likely to influence what happens can drastically alter your sense of control. For instance, the thought that an unexpected storm may compromise your hiking trip alters how much you feel you have control over your weekend’s plan. Across tasks as mundane as choosing a weekend activity or as consequential as planning a career, our sense of control shapes how we decide, how we learn, and how we feel.

Unsurprisingly, notions of ‘controllability’ and ‘empowerment’ have played a central role in both neuroscience and psychology^{1–6}, as well as in computer science and machine learning^{7–11}. From an evolutionary perspective, the desire for a sense of control has been described as a psychological and biological necessity¹, with far-reaching implications for well-being^{1–3} and developmental trajectories^{4,5}. In artificial intelligence, a parallel drive for control has inspired the development of agents capable of acquiring complex skills without external rewards or supervision^{7,8}. However, despite this convergence, there is currently no consensus on (i) what it means to be *objectively* in control of an environment, or (ii) when we, as humans, experience a *subjective* sense of control.

From an objective viewpoint, several engineering and mathematical measures of controllability have been proposed, each focusing on different environmental factors. Conventional control-theoretic measures—concerned with deterministic, continuous environments—quantify controllability in terms of the diversity, or more precisely, the dimensionality, of states that an agent can reach by selecting appropriate actions^{12,13}. Information-theoretic alternatives—more often concerned with discrete, stochastic environments—quantify a similar notion by measuring the diversity of reachable next states in terms of state entropy¹⁴. Yet other measures take a somewhat orthogonal approach and, using the notion of channel capacity¹⁵, quantify controllability by assessing how distinctly the agent’s available actions influence the environmental state^{16–18}.

At the same time, experimental quantities related to objective controllability have been shown to influence human behavioral^{19–23} and physiological responses^{24,25}. For instance, increasing the objective controllability of an environment can enhance motivation²⁶, increase happiness²⁷, and improve resilience to aversive events²⁸. These findings suggest that humans dynamically infer *some* quantities related to environmental controllability and adapt their behavior accordingly—in line with longstanding psychological theories^{29–32}. However, the relationships within and between objective and subjective notions of the sense of control remain underexplored. In particular, the absence of a clearly formalized, unifying notion of controllability has made it difficult to characterize the precise environmental factors that shape the human sense of control—or to describe crisply how these factors might differ across individuals.

Here, we offer a precise, yet interpretable, formulation of controllability as *a priori* preparedness to accomplish future tasks. This perspective unifies existing conceptions within a coherent, integrative framework and can be applied both to studying the human sense of control and to designing machine learning agents. We focus on the former and, across three theory-guided, large-scale experiments, demonstrate that our formulation accurately explains the human subjective sense of control. We show that this sense is shaped by the interplay between the availability of actions

and the possible and certain reachability of potentially desirable outcomes. Moreover, we show substantial variability in how individuals weigh these factors in their preferences. Thus, our work provides a solid foundation for future studies of controllability and the sense of control in both biological and artificial agents.

Results

We motivate our theoretical framework by developing our opening example into a thought experiment (Fig. 1A–B). Imagine your task is to pick an activity for the weekend, and you have four options in mind for where to go: a painting class, the circus, the theater, or a sports bar (Fig. 1A). Depending on where you live, only some of these activities may be accessible to you. For example, consider three cities, represented by c_1 , c_2 , and c_3 (Fig. 1A). In c_1 , all activities are available every weekend (Fig. 1A1), while in c_2 , only the painting class and theater are available (Fig. 1A2). In c_3 , by contrast, only the sports bar and circus are available, but on randomly chosen weekends, the circus venue hosts a painting class instead (Fig. 1A3). This means you cannot know in advance whether the painting class or the circus will be available on a given weekend. Given these scenarios, how much control would you feel in each city over choosing your weekend activity?

From an objective perspective—i.e., abstracting away from cognitive processing costs^{33,34} and the potential undesirability associated with an excessive number of options^{35,36}—it is reasonable to assume a maximal sense of control in c_1 , where all desired outcomes are reliably achievable; but how should we rank c_2 and c_3 ? To answer this question, we can model the thought experiment within a sequential decision-making framework^{37–39}, where your apartment in a given city is treated as the current state s , and each activity corresponds to a potential next state s' . Each city’s characteristics can then be summarized by a set of transition probabilities $p(s'|s, a)$, denoting the likelihood that an action a (arrows in Fig. 1A) takes you from your current state s to a subsequent state s' . Given this formalization, we can apply different controllability measures to rank the cities.

All existing mathematical measures assign the highest controllability to c_1 , consistent with our intuition about objective controllability. However, they diverge in how they rank c_2 and c_3 (Methods). For example, some measures, such as Klyubin empowerment¹⁶, assign equal control to c_2 and c_3 , since both offer two available actions with mutually exclusive outcomes (Fig. 1B2; see also Proposition 2). Other measures, such as maximum outcome entropy¹⁴, predict greater control in c_3 than in c_2 , because there is at least a chance of going to the circus if desired (Fig. 1B3). These differing predictions reflect fundamentally distinct perspectives on how to conceptualize and formalize objective controllability. Nevertheless, it remains unclear which perspective best matches our subjective sense of control.

We propose an integrative framework that unifies these different perspectives and enables them to be tested experimentally.

Sense of control as *a priori* preparedness

We begin with the intuitive assumption that the sense of control in a state s reflects the extent of *a priori* preparedness to reach any state s' that the agent may wish to reach. In other words, an agent perceives control when it feels prepared to reach any state s' as soon as it becomes desirable—where the desirability may stem from an internal decision or an external announcement. In the thought experiment of Fig. 1A, this perspective suggests that we feel a greater sense of control in

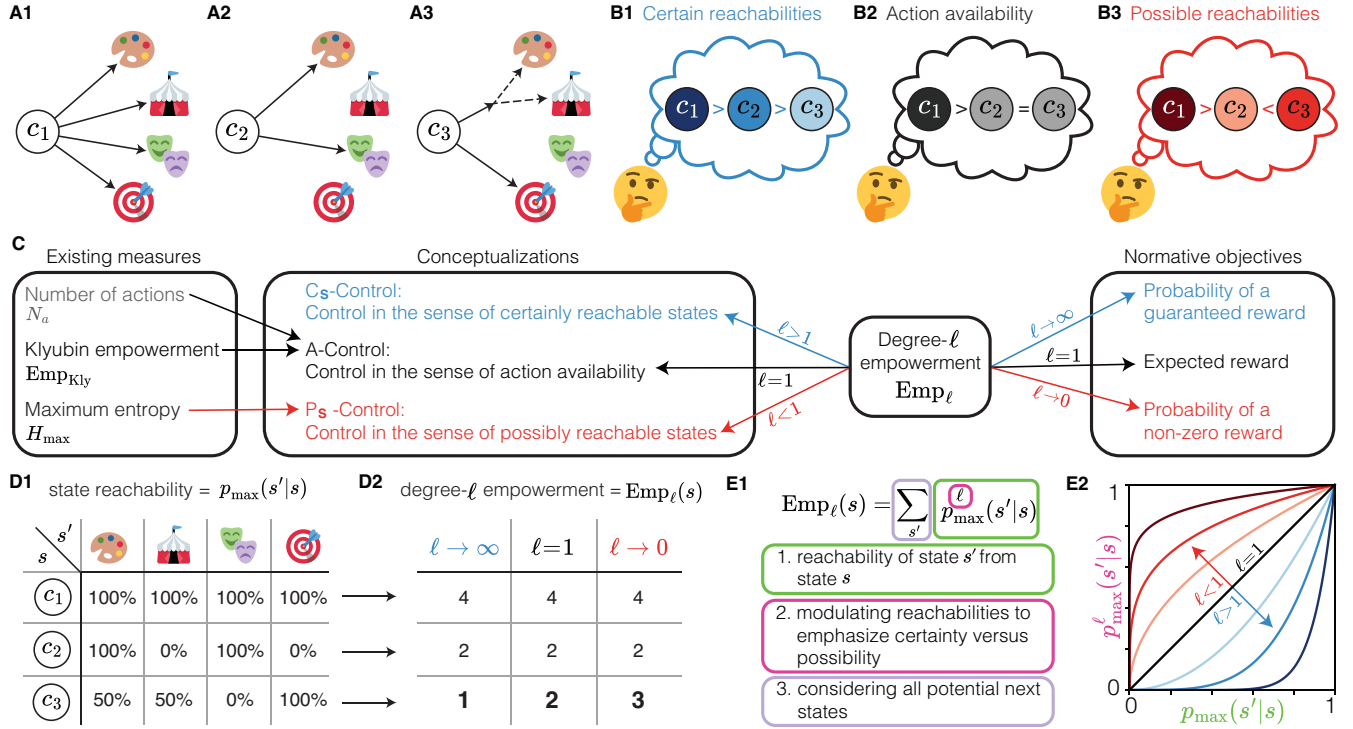


Figure 1: **An integrative framework for the sense of control.** **A.** Schematic of the thought experiment. We consider three cities as current states (c_1 , c_2 , and c_3) and four weekend activities (the painting class, the circus, the theater, and the sports bar) as potential next states. Solid arrows indicate available actions; dashed arrows indicate possible random transitions. **B.** Different conceptualizations of the sense of control yield different rankings of the cities shown in A; these conceptualizations can be directly linked to several existing measures of controllability (Methods). **C.** Framing the sense of control as ‘*a priori* preparedness for to-be-announced tasks’ bridges existing measures, distinct conceptualizations, and normative objectives via Emp_ℓ (Methods). ‘Number of actions’ is colored in gray to emphasize its simplicity in the sense that it does not depend on transition probabilities. **D.** For different choices of ℓ , Emp_ℓ quantifies the controllability of the cities in A in agreement with different views presented in B. **D1.** Reachability of each activity in the three cities. **D2.** Corresponding Emp_ℓ values for extreme choices of ℓ . Bolded values highlight how changes in ℓ affect the Emp_ℓ value assigned to c_3 . **E.** Although derived from an abstract, top-down perspective, Emp_ℓ has a straightforward interpretation. **E1.** For a given state s , $\text{Emp}_\ell(s)$ aggregates a modulated version of the reachabilities of all possible next states s' . **E2.** The modulation depends on the value of ℓ : Increasing (decreasing) ℓ downplays (amplifies) the contribution of lower reachabilities to Emp_ℓ . Credit: emojis adapted from ref.⁴⁰, under a CC BY 4.0 license.

a city where we feel more ‘prepared’ to pursue *any* activity we might want, despite not knowing in advance which activity it will be.

To formalize this idea, we assume that agents know (or believe they know) the actual dynamics of their environment (see Discussion); e.g., they know that the circus building sometimes hosts a painting class on random weekends. We also assume that there is no *a priori* extrinsic reward associated with specific environmental states; e.g., there is no *a priori* preference for any of the weekend activities. Then, at some later point, a goal state g is randomly selected and revealed to the agent; for instance, going to the circus is chosen as the preferred activity for that weekend. The key question then becomes: ‘In which state is an agent best prepared to reach g , once it has been announced?’

This problem can be framed within the framework of sequential decision-making^{37–39} by assigning a reward of 1 to the goal state g and 0 to all other states. In this setting, the question of ‘Where would an agent be most prepared?’ depends on which reward statistics the agent seeks to

optimize (i.e., its objective; Fig. 1C). We show that preparedness with respect to different reward statistics can be formally translated into preferences for distinct environmental features that are directly linked to controllability (see Methods). For example, if the agent aims to maximize expected reward, it prefers states with many available *distinct* actions (A-Control; see Proposition 1 and Corollary 1–2), where actions are considered distinct if they lead to different next states. Specifically, the expected reward corresponds to a regularized version of the mere number N_a of actions, where the regularization term captures similarities between actions based on their transition probabilities (Corollary 1): The higher the similarity, the lower the number of effectively distinct actions. On the other hand, if the goal is to maximize the probability of receiving a guaranteed reward, the agent favors states with many *certainly* reachable next states (C_S-Control; see Methods), whereas if the objective is to *minimize* the chance of *not* obtaining a reward, it prefers states with many *possibly* reachable next states (P_S-Control; see Methods).

Accordingly, the different normative objectives correspond to distinct conceptualizations of the sense of control that are all intuitively sensible yet fundamentally different (Fig. 1C). Our thought experiment further illustrates these distinctions (Fig. 1B). A-Control predicts that the agent experiences the same degree of control in c_2 and c_3 (Fig. 1B2). Hence, A-Control is consistent with Klyubin empowerment¹⁶ which, in this example, counts the mere number N_a of actions, as all actions are distinct. In contrast, P_S-Control predicts greater perceived control in c_3 than in c_2 (Fig. 1B3), aligning with the maximum outcome entropy¹⁴. As the example makes clear, it is crucial not to conflate the number of *available actions* (captured by A-Control) with the number of *reachable outcomes* (captured by P_S-Control). In contrast to A-Control and P_S-Control, C_S-Control predicts a higher sense of control in c_2 than in c_3 (Fig. 1B1).

In conclusion, these theoretical results establish a formal connection between performance objectives in value-based decision-making and the constructs of control and empowerment, two topics that are often treated as conceptually independent.

Emp_ℓ unifies and expands existing perspectives on the sense of control

Critically, we found that the three normative objectives correspond to special cases of a single, interpretable measure that we call degree- ℓ empowerment, Emp_ℓ (Methods; Fig. 1C–E). Intuitively, Emp_ℓ quantifies the sense of control in a state s as how ‘easily,’ on average, the agent can transition to other states in the environment, if it wishes to do so (Fig. 1E1):

$$\text{Emp}_\ell(s) := \sum_{s'} p_{\max}^\ell(s'|s), \quad (1)$$

where $p_{\max}(s'|s)$ denotes the maximum probability $\max_a p(s'|s, a)$ of reaching s' from s in a single step. We refer to $p_{\max}(s'|s)$ as the (single-step) *reachability* of s' from s (see the Discussion for potential generalizations to the multi-step case). The degree ℓ is a free parameter that modulates the contribution of each reachability (Fig. 1E2). For $\ell > 1$, increasing ℓ progressively downplays the influence of small reachabilities. As a result, in the limit as $\ell \rightarrow \infty$, Emp_ℓ counts only *certain* reachabilities ($p_{\max}(s'|s) = 1$). For $\ell < 1$, decreasing ℓ progressively exaggerates the influence of small reachabilities. In the limit as $\ell \rightarrow 0$, Emp_ℓ counts any *possible* reachability ($p_{\max}(s'|s) > 0$), regardless of how small. At the critical value $\ell = 1$, Emp_ℓ(s) is proportional to the expected future reward and captures the distinctiveness of available actions in state s (Methods).

Consequently, Emp_ℓ connects and encompasses the distinct conceptualizations of the sense of con-

trol (Fig. 1C). Our thought experiment clearly illustrates this integration: Varying ℓ leads to different predictions from Emp_ℓ , aligning with perspectives that previously appeared contradictory (compare Fig. 1B and Fig. 1D). In addition, Emp_ℓ satisfies several theoretically desirable properties that one would intuitively expect from an appropriate measure of the sense of control (Theorem 1–2, and Proposition 3). In particular, Emp_ℓ aligns with psychological theory of self-efficacy^{29,31}. Following the original conceptualization of self-efficacy³¹, we model reduced self-efficacy as action-execution noise (Methods) and show that increasing this noise directly decreases the sense of control as measured by $\text{Emp}_\ell(s)$ for any ℓ (Theorem 2). Thus, self-efficacy theory is consistent with all three conceptualizations of control in Fig. 1B–C. Importantly, Emp_ℓ captures how reduced self-efficacy diminishes perceived controllability even when the true transition probabilities of the environment remain unchanged.

In conclusion, Emp_ℓ both *unifies* and *expands* existing perspectives on the sense of control, providing a principled foundation for investigating how humans experience control. Using this foundation, we performed three experiments to investigate whether Emp_ℓ captures different operationalizations of the human sense of control.

Experiment 1: *A priori* preparedness to collect gold

As the first experiment, we designed a paradigm that closely resembled our thought experiment in Fig. 1A. The goal of the experiment was to determine which states participants preferred when they were unaware of their future task. The underlying idea was to treat these preferences as an indirect operationalization of participants’ sense of control.

The experiment was divided into two phases (Fig. 2A). In the first phase, 71 participants played 10 rounds of a gold-collecting game (Fig. 2B). In each round, participants began at the center of a 3×3 virtual room and waited for a gold coin to randomly appear in one of the peripheral states (Fig. 2B). They then selected one (and only one) available action to reach the gold coin. If an action had a single endpoint (indicated by green disks in Fig. 2B), it deterministically moved the participant to that endpoint. If the action had multiple endpoints (e.g., as in Room 2 in Fig. 2C), it moved the participant to one of these endpoints at random, with equal probability.

In the second phase, participants were shown 132 pairs of rooms and asked to choose which room they would prefer for collecting gold coins (Fig. 2C). The experiment included 12 prespecified rooms that varied along two dimensions: the number of available actions (2, 3, 4, or 8) and the level of outcome stochasticity associated with each action (see Supplementary Materials for the precise characteristics of the rooms). Depending on the room, the gold coin could appear in a location that was deterministically reachable or unreachable (Fig. 2D1), or in a location that could be only probabilistically reachable (Fig. 2D2). This design allowed us to make trials in which comparing different rooms resembled comparing different cities in the thought experiment. For example, Room 1 and Room 2 in Fig. 2C qualitatively mirror the properties of c_2 and c_3 in Fig. 1A. In this way, we could test whether participants’ preferences reflect C_S-Control (i.e., preferring Room 1 over Room 2), P_S-Control (i.e., preferring Room 2 over Room 1), or A-Control (i.e., showing no systematic preference between the two rooms).

Importantly, in the second phase, participants were only asked to choose between two rooms; they were informed that the experimenter would play the gold-collecting game on their behalf by *always* selecting the most appropriate action. They were told that their monetary bonus would increase each time a gold coin was successfully collected for them (Fig. 2A). They received feedback on the

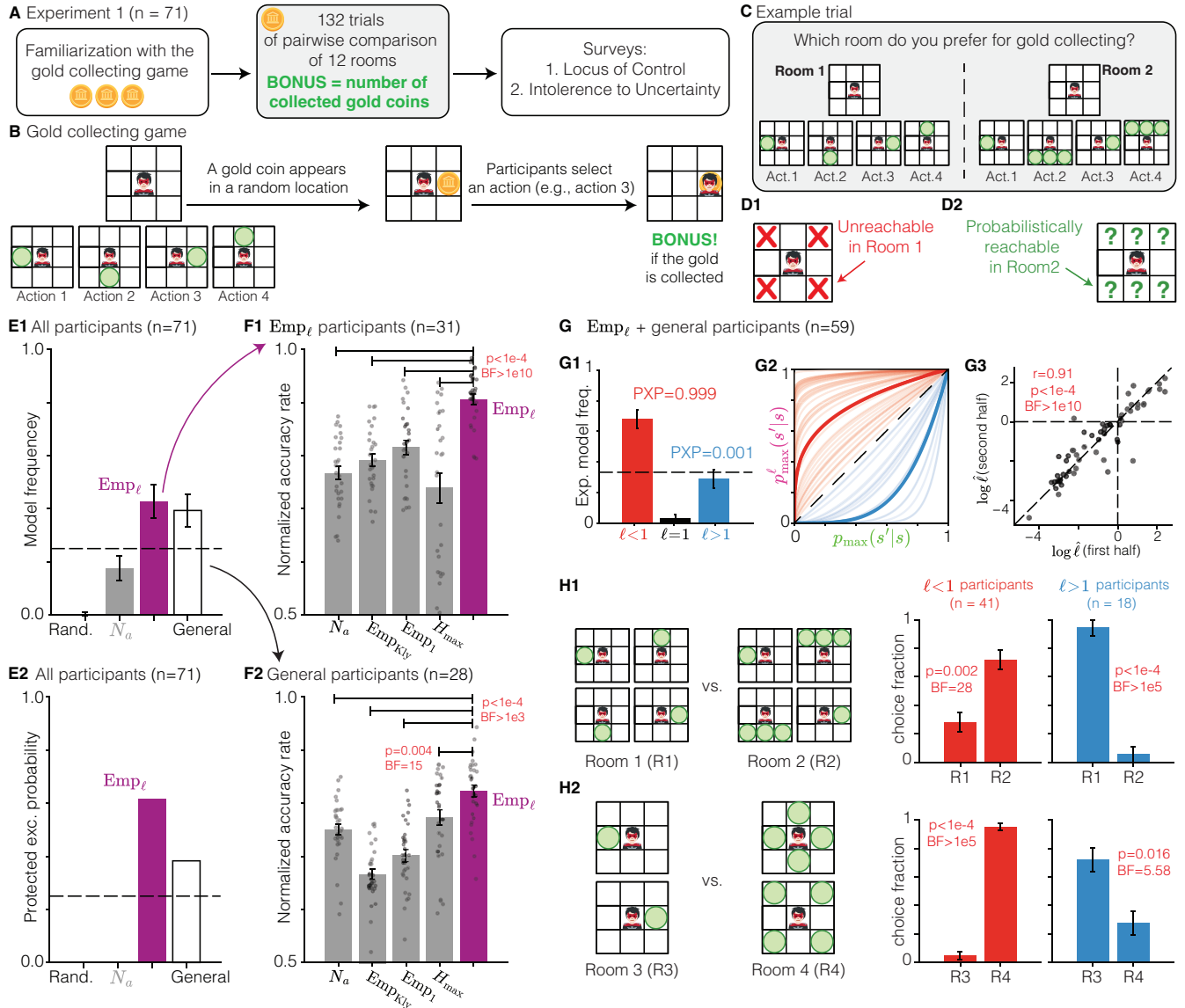


Figure 2: Experiment 1: Emp_ℓ accurately captures individual differences in preference for certainty versus possibility. **A.** Experiment 1. Participants first completed a familiarization phase with the gold-collecting game (see **B**), followed by 132 room-selection trials (see **D**) and, finally, the LOC⁴¹ and IUS⁴² surveys. **B.** Gold-collecting game. Participants waited in the center of a 3×3 virtual room until a gold coin appeared in one of the peripheral states. They then selected a single action to reach and collect the coin. **C.** In each room-selection trial, participants chose which of the two presented rooms they preferred for collecting gold coins. **D.** Rooms differed in the certainty and possibility of reaching the gold coin. Depending on the room, some locations were entirely unreachable (e.g., Room 1 in **C**; **D1**) or only probabilistically reachable (e.g., Room 2 in **C**; **D2**). **E.** Model comparison of participants' preferences under (i) a random model, (ii) N_a , (iii) Emp_ℓ , and (iv) the general Bradley-Terry model^{43,44} (Methods). **E1.** Model frequencies indicate the proportion of participants best explained by each model. Error bars: posterior standard deviations (SD). **E2.** Protected exceedance probabilities (PXP)⁴⁵ indicate the probability that each model is more frequent than all alternatives. **F.** Normalized, cross-validated accuracy of Emp_ℓ and the alternative measures (Fig. 1) in predicting human choices. Emp_ℓ outperforms all alternatives in both the Emp_ℓ (**F1**) and general (**F2**) participants. Error bars: the standard error of the mean (SEM). Points: individual participants. **G.** Individual differences in possibility-certainty preferences of the Emp_ℓ and general participants. **G1.** Model frequencies for $\ell < 1$, $\ell = 1$, and $\ell > 1$. **G2.** Variations in reachability modulation (see Fig. 1E2), quantified by the posterior expectation $\log \hat{\ell}$; thick lines show group means. **G3** Posterior expectations of $\log \hat{\ell}$ estimated separately from the first and second halves of the data. **H.** Fraction of choices of different participant groups favoring Room 1 over Room 2 (**H1**) and Room 3 over Room 4 (**H2**); see Supplementary Materials for data on comparing other rooms. Error bars: the SEM. Red p-values: statistically significant effects (FDR controlled at 0.05)⁴⁶. Red Bayes Factors (BF): strong evidence for the alternative hypothesis ($BF \geq 3$). Credit: emojis adapted from ref.⁴⁰, under a CC BY 4.0 license.

total number of coins collected only once every seven trials; we did not observe any temporary change in their behavior in the trials following the feedback (Supplementary Materials). At the end of the experiment, participants completed two questionnaires: the Locus of Control (LOC) scale⁴¹ and the Intolerance of Uncertainty (IUS) scale⁴². The LOC survey was administered to examine whether potential differences in participants’ preferences for P_S-Control, C_S-Control, or A-Control are associated with how they attribute everyday life events to themselves versus to chance. Similarly, the IUS survey was used to assess whether these preferences are related to individual differences in intolerance of risk and uncertainty. No additional questionnaire was included in order to keep the session brief.

Emp_ℓ accurately predicts human behavior

We used Bayesian model selection^{45,47} to compare three hypotheses regarding participants’ room preferences (**Methods**). The first hypothesis assumed that participants ignore how the actions’ endpoints are arranged and simply prefer rooms with a higher number of available actions (N_a ; e.g., see ref.²⁷). The second hypothesis proposed that participants prefer the room with the highest Emp_ℓ value (to be precise, the Emp_ℓ value of the room’s central state); we treated ℓ as a participant-specific free parameter. The third and most flexible hypothesis modeled participants’ choices using the general Bradley–Terry model^{43,44}, in which each of the 12 rooms was assigned a participant-specific subjective utility parameter. This model subsumes any self-consistent preference structure, including strategies based on N_a , Emp_ℓ, other controllability measures, or potential heuristics; however, the model’s added flexibility was penalized through Bayesian model selection^{48,49} (see Supplementary Materials for model-recovery). Finally, we also included a random null model to validate that our exclusion criteria effectively removed inattentive or randomly behaving participants (**Methods**, Supplementary Materials).

The choices of the majority of participants were best explained by either Emp_ℓ ($n = 31$) or the general Bradley-Terry model ($n = 28$; **Fig. 2E** and Supplementary Materials). The assignment of the largest group of participants to Emp_ℓ is particularly notable given that the general Bradley-Terry model was available as an alternative: It rules out the possibility that an untested measure of sense of control could substantially outperform Emp_ℓ in explaining these participants’ choices. We refer to these 31 participants as ‘Emp_ℓ participants’ and to the 28 participants best explained by the general Bradley-Terry model as ‘general participants.’

The worse performance of Emp_ℓ relative to the general Bradley-Terry model among general participants could have several explanations; for example, there may exist another controllability measure that could have better explained their behavior, or different participants might have relied on idiosyncratic heuristics that could have been only captured by the flexibility of the general Bradley-Terry model. To assess these possibilities, we evaluated the predictive accuracy of Emp_ℓ using two-fold cross-validation⁵⁰ (**Methods**; **Fig. 2F**) and compared it with that of the representative measures of environmental controllability reviewed in **Fig. 1**.

For the 31 Emp_ℓ participants, Emp_ℓ achieved an average normalized accuracy of more than 90% (**Fig. 2F1**) and outperformed every alternative model by at least 10% on average. Importantly, among the 28 general participants, Emp_ℓ still achieved an average normalized accuracy of more than 80% and significantly outperformed all existing alternative models by at least 5% (**Fig. 2F2**). Thus, the statistical advantage of the general Bradley-Terry model in the Bayesian model comparison cannot be attributed to another existing, representative measure of controllability explaining

general participants’ choices better than Emp_ℓ . Additional analyses suggested that the choices of general participants may reflect heuristic approximations to Emp_ℓ , including a tendency to assign lower values to actions whose endpoints lie in corner locations (see Supplementary Materials).

Together, these results suggest Emp_ℓ as a highly accurate model of human preferences in our task, albeit an approximate one for some participants.

Emp_ℓ captures individual preferences for certainty versus possibility

We next pooled the Emp_ℓ and general participants and analyzed the posterior expectation $\hat{\ell}$ to determine which conceptualization of the sense of control (Fig. 1C) best aligned with their preferences (see Supplementary Materials for parameter recovery⁴⁹). We observed substantial individual differences (Fig. 2G-H), with promising split-half consistency (Fig. 2G3). Approximately two-thirds of participants preferred *possible* reachability, whereas one-third preferred *certain* reachability (Fig. 2G1). Specifically, participants preferring possibility consistently chose rooms with the highest number of *possibly* reachable locations, even when these locations were only probabilistically reachable (e.g., Rooms 2 and 4 in Fig. 2H; see Supplementary Materials). In contrast, the other group consistently chose rooms with the highest number of *certainly* reachable locations, even at the expense of some locations being definitively unreachable (e.g., Rooms 1 and 3 in Fig. 2H; see Supplementary Materials). Within each group, we additionally observed considerable variability in the strength of these preferences (Fig. 2G2 and Fig. 1E2). Nonetheless, the group-averaged behavior closely matched the qualitative predictions of the corresponding conceptualizations (Fig. 2H and Supplementary Materials).

In conclusion, we found strong quantitative (Fig. 2G) and qualitative (Fig. 2H and Supplementary Materials) evidence for individual differences in participants’ preferences for certainty versus possibility. While Emp_ℓ effectively captures these differences, existing measures of the sense of control do not account for this certainty–possibility dichotomy.

Experiment 2: *A priori* preparedness to collect gold or avoid bombs

We derived Emp_ℓ from the premise that the sense of control reflects *a priori* preparedness to reach a desired goal state that has yet to be announced. In Experiment 1, we observed that Emp_ℓ either fully captures or closely approximates the corresponding operationalization of control in humans. However, the formal correspondence between Emp_ℓ and different normative objectives (Fig. 1C) holds for any reward distribution that does not impose *a priori* preferences over specific subsets of states (Theorem 1). This correspondence becomes particularly informative in the loss domain.

Specifically, extensive research in value-based decision-making shows that human risk preferences systematically differ between gains and losses, even when the underlying choice structure is formally equivalent^{51–53}. The rooms in Experiment 1 were designed to entail different degrees of risk in collecting the gold. However, because the reward was binary, a classical account of risk in terms of reward variance is insensitive to the certainty–possibility distinction between rooms: Room 3 and Room 4 in Fig. 2H2, for example, both yield an overall 25% probability of an identical reward. If, instead, we isolate the ‘risk’ arising from uncertainty about the gold’s location, then a room is riskier the more its locations vary in reachability: A 25% chance of obtaining the gold wherever it happens to be (Room 4 in Fig. 2H2) entails less ‘risk’ than a guaranteed reward at only 25% of possible locations (Room 3 in Fig. 2H2). Thus, the stronger preference for possibility in Experiment 1 may

be partially explained by the well-established tendency toward risk aversion in the gain domain⁵¹. If this interpretation is correct, and the individual differences in certainty–possibility preferences observed in Experiment 1 merely reflect the risk attitudes studied in value-based decision-making, then these preferences should reverse or shift under a loss framing, with most participants instead preferring certainty.

To test this, we conducted a second experiment in which we examined whether participants’ preferences for certainty versus possibility would shift when the task was framed as avoiding punishments rather than collecting rewards ($n = 148$; Fig. 3A–C). Specifically, we designed loss-based trials that exactly mirrored the gain-based ones by assigning an equal punishment (i.e., a ‘reward’ of -1) to all states except a yet-to-be-designated safe state g . In this case, Emp_ℓ reflects the *a priori* preparedness to *avoid* all states except the yet-to-be-announced g (Corollary 3).

The experiment included a gold block (representing the gain domain) and a bomb block (representing the loss domain), where the block order was randomized across participants (Fig. 3A). Specifically, 71 participants completed the gold block first, thereby replicating Experiment 1 in this part, and 77 participants completed the bomb block first. The procedure for the gold block was identical to Experiment 1 (Fig. 2A–D), with one minor difference: Only 9 of the 12 prespecified rooms were used in Experiment 2 to shorten the total experiment time (Supplementary Materials). This resulted in 72 room-selection trials per block (Methods), for a total of 144 across both blocks. The bomb block followed the same overall structure as the gold block, with one key difference: During the familiarization phase, participants played 10 rounds of a bomb-avoiding game (Fig. 3B), and in the room selection phase, they chose which room they would prefer for avoiding bombs (Fig. 3C). In each round of the bomb avoiding game, participants began in the center of a 3×3 virtual room filled with bombs and waited until one of the eight bombs randomly disappeared (Fig. 3B). They then selected one (and only one) available action to move to the newly revealed safe location.

As in Experiment 1, participants were only asked to choose between pairs of rooms. They were informed that the experimenter would play both the gold-collecting and bomb-avoiding games on their behalf by always selecting the most appropriate action. In the bomb block, participants were told that they would lose one gold coin each time we failed to escape the bombs for them. They received feedback on the total number of lost coins only once every seven trials; we did not observe any temporary change in their behavior in the trials following the feedback (Supplementary Materials). To ensure that no participant went into debt, we provided each participant with 50 extra gold coins at the beginning of each bomb block (Methods). At the end of the experiment, all participants completed the LOC and IUS surveys.

Emp_ℓ accurately predicts human behavior in both gold and bomb trials

We repeated the Bayesian model-selection procedure used in Fig. 2E, applied separately to the gold and bomb trials. For both trial types, the choices of the largest group of the 148 participants were best explained by Emp_ℓ , followed by the general Bradley–Terry and the N_a models (Fig. 3D). The expected model frequencies were highly similar across gold and bomb trials (Fig. 3D1) and approximately matched those observed in Experiment 1 (Fig. 2E). Consistent with these results, Emp_ℓ achieves an approximately 90% normalized accuracy rate for the Emp_ℓ participants (Fig. 3E1), and 80% for the general participants (Fig. 3E2), outperforming all existing, representative measures of controllability. Together, these findings demonstrate that the core results from Experiment 1 in

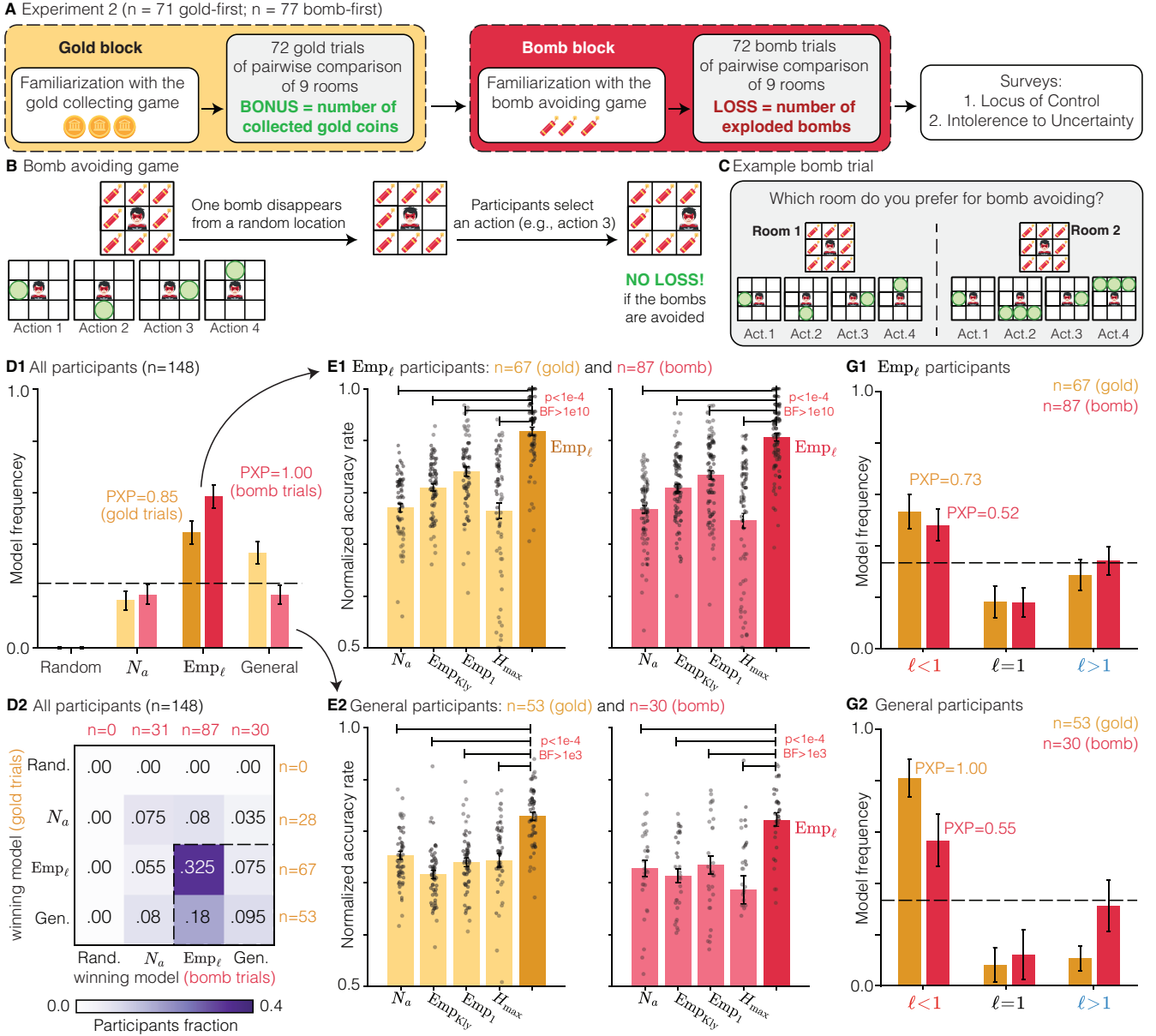


Figure 3: Experiment 2: Emp_ℓ accurately predicts human behavior in both gain and loss domains. **A.** Experiment 2 included a gold block and a bomb block. The gold block replicated the first two phases of Experiment 1 (Fig. 2A), using only 9 of the 12 predefined rooms (Supplementary Materials). The bomb block followed the same structure but focused on bomb avoidance rather than gold collection. Block order was randomized across participants and did not influence the results (Supplementary Materials). Participants completed the LOC⁴¹ and IUS⁴² surveys at the end of the experiment. **B.** Bomb avoiding game. Participants waited in the center of a 3×3 virtual room filled with bombs until one of the bombs randomly disappeared. They then selected a single action to reach the resulting safe location. **C.** In room-selection trials, participants chose the room they would prefer for the bomb avoiding game. **D.** Model comparison results analogous to Fig. 2E. **D1.** Model frequencies, separately for the gold trials (yellow) and bomb trials (red). **D2.** Fraction of participants best explained by each model. **E.** Normalized, cross-validated accuracy of Emp_ℓ and the alternative measures (Fig. 1) in predicting human choices, separately for the gold trials (left, yellow) and bomb trials (right, red). Emp_ℓ outperforms all alternatives in both the Emp_ℓ (**F1**) and general (**F2**) participants. **G.** Model frequencies comparing $\ell < 1$, $\ell = 1$, and $\ell > 1$ (see Fig. 1C), separately for the gold trials (yellow) and bomb trials (red). This analysis includes participants whose preferences were best explained by either Emp_ℓ (**G1**) or the general Bradley-Terry model (**G1**). Same format as Fig. 2G1. Red p-values: statistically significant effects (FDR controlled at 0.05)⁴⁶. Red BF: Strong evidence for the alternative hypothesis ($\text{BF} \geq 3$). Error bars: posterior SD. *Credit: emojis adapted from ref.⁴⁰, under a CC BY 4.0 license.*

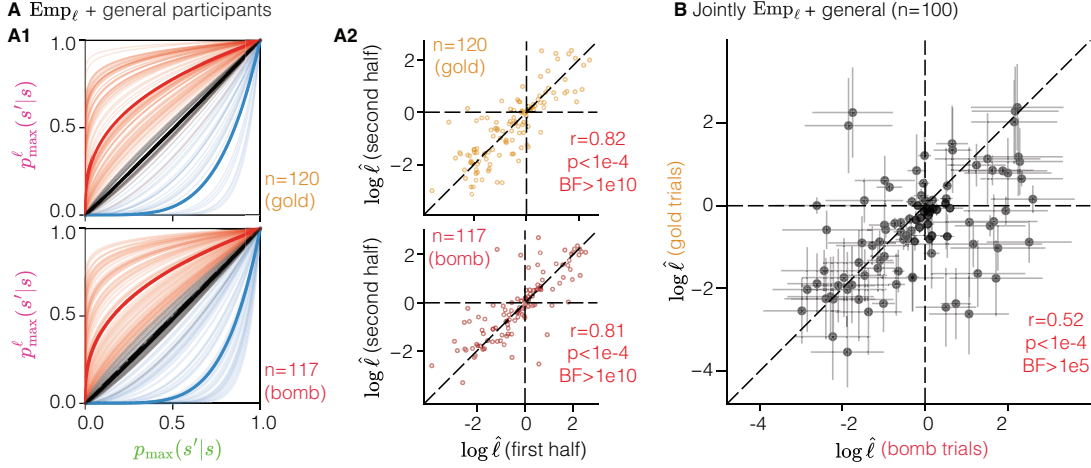


Figure 4: **Experiment 2: Individual differences in preference for certainty versus possibility are correlated across gain and loss domains.** **A. A1.** Variations in reachability modulation (see Fig. 1E2), quantified by the posterior expectation $\log \hat{\ell}$, separately for the gold block (top) and bomb block (bottom). Thick lines show group means. **A2.** Posterior expected values of $\log \hat{\ell}$ evaluated separately for the first and second halves of the data, separately for the gold block (top) and bomb block (bottom). **B.** Relationship between the posterior expected values of $\log \hat{\ell}$ in the gold and bomb trials. This analysis includes only participants whose preferences were best explained by Emp_{ℓ} or the general Bradley-Terry model in both blocks (the marked four cells in Fig. 3D2).

the reward-seeking context generalize robustly to the punishment-avoiding context.

However, we observed a slightly higher model frequency for Emp_{ℓ} in the bomb trials than the gold trials (exceedance probability > 0.95 ; Fig. 3D1). Specifically, Emp_{ℓ} best explained the preferences of 32% of participants jointly in both trial types (Fig. 3D2). However, whereas an additional 26% of participants showed Emp_{ℓ} -consistent preferences exclusively in the bomb trials, only 12% of participants showed Emp_{ℓ} -consistent preferences exclusively in the gold trials (Fig. 3D2). This pattern suggests that humans may be slightly more inclined to follow Emp_{ℓ} in punishment-avoiding contexts than in reward-seeking ones. We found no substantial difference between participants according to whether they started with the gold or bomb block (Supplementary Materials).

Certainty–possibility preferences are correlated across gold and bomb trials

We next analyzed $\hat{\ell}$ separately for the Emp_{ℓ} and general participants in the gold and bomb trials. As in Experiment 1, we observed substantial individual differences (Fig. 3G and Fig. 4A1), along with promising split-half consistency (Fig. 4A2). For the Emp_{ℓ} participants, the overall distribution of participants’ preferences for certainty versus possibility was similar across trial types and consistent with Experiment 1 (Fig. 3G1 versus Fig. 2F1). Nevertheless, for the general participants, we observed a generally higher preference for possibility than certainty in the gold trials, but this tendency was slightly lower in the bomb trials (exceedance probability > 0.95 ; Fig. 3G2). This modest shift toward certainty in the loss domain aligns qualitatively with findings from behavioral economics⁵¹, though the effect was notably smaller than might be expected and only limited to the general participants. Crucially, among participants whose preferences were best explained by either Emp_{ℓ} or the general Bradley-Terry model in *both* trial types, we found a strong positive correlation between their $\hat{\ell}$ values in the gold and bomb blocks (Fig. 4B).

In conclusion, these findings demonstrate that (i) preferences for certainty versus possibility remain relatively stable across both reward-seeking and punishment-avoiding contexts, and (ii) Emp_ℓ reliably captures these individual differences. These provide strong evidence that individual differences in certainty–possibility preferences cannot be simply equated with risk attitudes in value-based decision-making.

Experiment 3: Explicit control-judgment

In the first two experiments, we showed that Emp_ℓ captures two indirect operationalizations of the human sense of control. These operationalizations were grounded in our theoretical results that formally link preparedness to environmental controllability (Fig. 1). However, because both experiments relied on an instrumental framing (i.e., choosing rooms to collect gold or avoid bombs), we could not fully exclude the possibility that participants’ preferences were additionally driven by value-based decision-making factors or heuristics beyond perceived controllability. Although Experiment 2 provided evidence against a simple risk-based account of room preferences, this evidence remained indirect.

We therefore conducted a third, pre-registered experiment to test directly whether Emp_ℓ captures humans’ explicit control judgments ($n = 174$; Fig. 5). The overall structure closely mirrored Experiment 1 (Fig. 2A versus Fig. 5A), with the key difference that the room-selection phase probed in which room participants would feel *more in control*, rather than which room they preferred for collecting gold (Fig. 2C versus Fig. 5B). The design was inspired by prior control-judgment paradigms^{54–58}, but emphasized relative comparisons over absolute ratings⁵⁹.

The experiment was divided into two phases (Fig. 5A). In the first phase, participants completed 10 rounds of the gold-collecting game. Unlike the first two experiments, this phase was introduced as the main task, with participants’ actions directly influencing their bonus. From the experimenters’ perspective, the primary purpose of this phase was to create an engaging setup that allowed participants to gain direct experience of the transition structure of the different rooms. The second phase was presented as an additional, post-experiment phase independent of gold collection. Participants were informed that their bonus from the first phase was secure and that they could earn an additional one-off performance-based bonus in the second phase, contingent on passing attention checks (Methods).

Participants then completed 72 control-judgment trials (pairwise comparisons of the same 9 rooms as in Experiment 2), indicating the room in which they would feel more in control (Fig. 5B). They were explicitly informed that there was no objectively correct answer, emphasizing subjective judgments rather than performance optimization (Methods). At the end of the experiment, all participants completed the LOC and IUS surveys.

Emp_ℓ accurately predicts human explicit control-judgment

Given the structural equivalence of Experiments 1 and 3, we repeated all analyses from Experiment 1 for the control-judgment trials in Experiment 3 (Fig. 2 versus Fig. 5; see also Supplementary Materials). Overall, control judgments in Experiment 3 qualitatively mirrored the room-selection choices for gold collecting in Experiment 1 but additionally provided stronger quantitative evidence in favor of Emp_ℓ . Specifically, the choices of close to 80% of participants were best explained by Emp_ℓ (Fig. 5C), with Emp_ℓ achieving a normalized accuracy of nearly 95% for Emp_ℓ participants

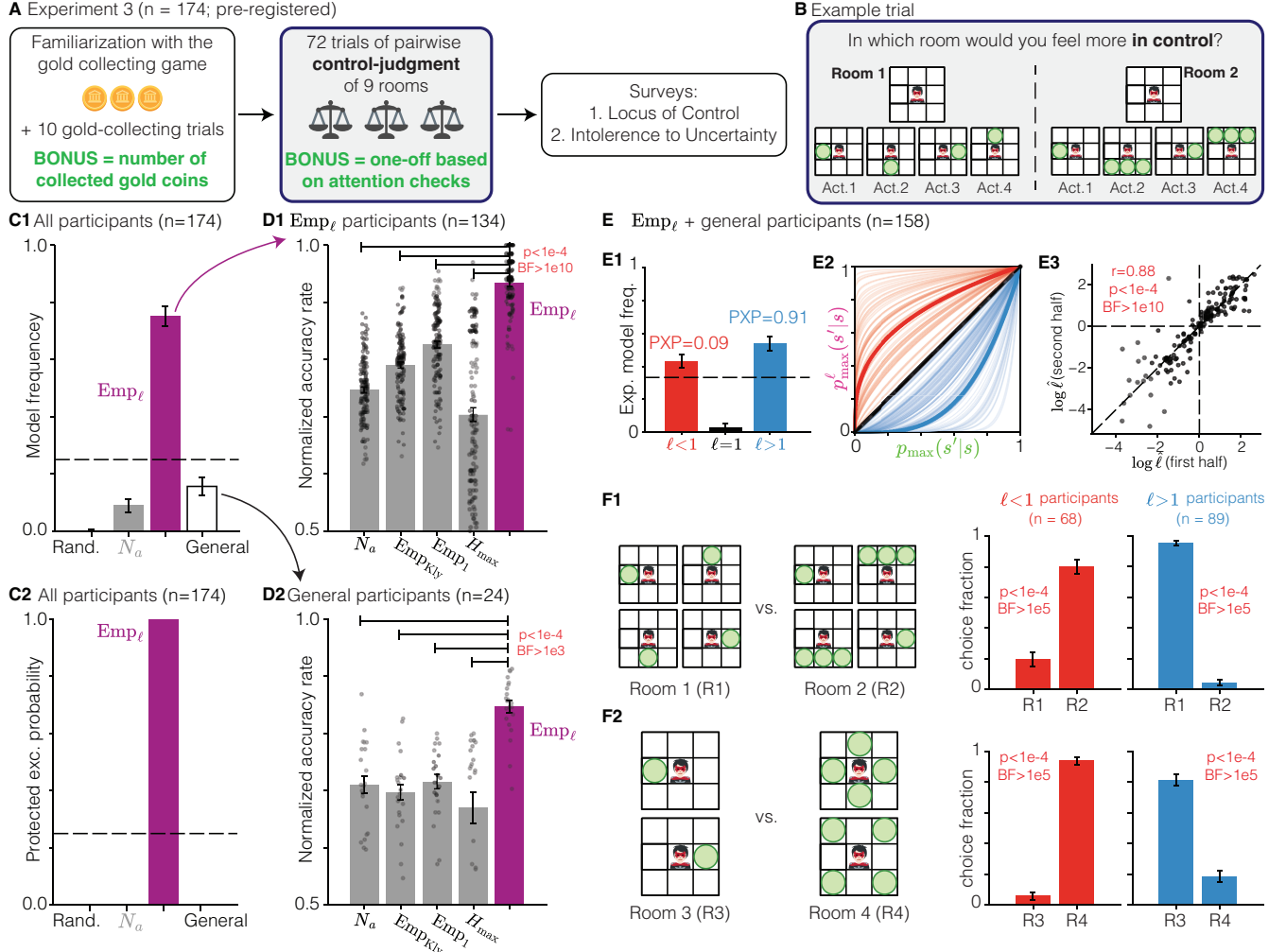


Figure 5: Experiment 3: Certainty–possibility differences captured by Emp_ℓ closely align with human explicit control judgments. **A.** Participants first completed a familiarization phase with the gold-collecting game, followed by 10 gold-collecting trials that determined their monetary bonus. They were then informed that the second part of the experiment was independent of gold collecting, with a one-off bonus contingent on passing attention checks (Methods). Participants then completed 72 control-judgment trials (see B; using the same 9 prespecified rooms as in Experiment 2), followed by the LOC⁴¹ and IUS⁴² surveys. **B.** In each control-judgment trial, participants chose the room in which they would feel more in control. **C.** Model comparison results, analogous to Fig. 2E, showing model frequencies (C1) and PXP (C2). Error bars: posterior SD. **D.** Normalized, cross-validated accuracy rate, analogous to Fig. 2F for both the Emp_ℓ (D1) and general (D2) participants. Error bars: the SEM. Points: individual participants. **E.** Individual differences in possibility–certainty preferences of the Emp_ℓ and general participants. **E1.** Model frequencies for $\ell < 1$, $\ell = 1$, and $\ell > 1$. **E2.** Variations in reachability modulation (see Fig. 1E2), quantified by the posterior expectation $\log \hat{\ell}$; thick lines show group means. **E3** Posterior expectations of $\log \hat{\ell}$ estimated separately from the first and second halves of the data. **F.** Fraction of choices of different participant groups favoring Room 1 over Room 2 (F1) and Room 3 over Room 4 (F2); see Supplementary Materials for data on comparing other rooms. Error bars: the SEM. Red p-values: statistically significant effects (FDR controlled at 0.05)⁴⁶. Red Bayes Factors (BF): strong evidence for the alternative hypothesis ($BF \geq 3$). Credit: emojis adapted from ref.⁴⁰, under a CC BY 4.0 license.

(Fig. 5D1) and 85% for the general participants (Fig. 5D2). Similar to the first two experiments, we observed substantial individual differences in how participants weighed certainty versus possibility in their control judgments (Fig. 5E-F and Supplementary Materials), with promising split-half consistency (Fig. 5E3).

Intriguingly, there were two main differences between the results of Experiments 1 and 3. First, fewer than 20% of participants were assigned to the general Bradley-Terry model in Experiment 3, as opposed to the $\sim 40\%$ frequency in Experiment 1 (Fig. 2E1 versus Fig. 5C1). This substantial decrease provides evidence that human subjective control judgments closely mirror Emp_ℓ , with at most very modest influence from other possible controllability measures or potential heuristics.

Second, in Experiment 3, the proportions of certainty- and possibility-preferring participants were nearly equal in control judgments (Fig. 5D1), whereas in Experiment 1 the balance favored possibility-preferring in gold collecting by almost 30% (Fig. 2G1). This shift is consistent with the slight decrease in the proportion of possibility-preferring participants in the bomb trials of Experiment 2 relative to the gold trials (Fig. 3G), suggesting that participants’ choices in Experiments 1 and 2 may have been mildly influenced by risk preferences⁵¹, given the instrumental framing of these tasks. Nevertheless, the substantially higher model frequency and normalized accuracy of Emp_ℓ in Experiment 3 than in Experiments 1 and 2 suggest that Emp_ℓ captures a property of the rooms that closely aligns with human judgments of controllability, rather than merely reflecting a risk-seeking component of decision-making.

Together, the three experiments show that Emp_ℓ successfully captures the human sense of control across three distinct framings and operationalizations.

Origins of individual differences in certainty–possibility preferences

Finally, we investigated potential sources of the observed individual differences in preferences for certainty versus possibility. One hypothesis is that these differences might be rooted in specific personality traits. To test this, we examined correlations between participants’ $\hat{\ell}$ values and their responses to the LOC and IUS surveys (36 items in total). Across all three experiments, we found no consistent, significant correlation between $\log \hat{\ell}$ and the raw survey scores (Supplementary Materials). We then conducted a Principal Component Analysis (PCA) on the survey responses and identified the main principal components (Supplementary Materials). However, none of these components consistently correlated with $\log \hat{\ell}$ across the three experiments. These findings suggest that the traits measured by the LOC and IUS surveys may be distinct from the underlying factors driving participants’ behavioral preferences in our paradigm.

Discussion

From planning a weekend to navigating major life decisions, our experiences are shaped by how much we feel in control of our future. Such observations have long been motivated psychological theories of learned helplessness^{30,60,61}, self-efficacy³¹, agency^{2,62,63}, and autonomy⁶⁴. These theories emphasize how people *subjectively* perceive and adapt to *objective* controllability of their environment, which in turn has been formalized by computer scientists and engineers^{12,13,16}. In the meantime, the objective measures have been successfully used to design complex artificial agents that flexibly manipulate their environments^{7–9,65}. Yet the lack of a formal, unifying framework has

made it challenging to identify which environmental factors shape the human sense of control or are critical for artificial intelligence.

In this work, we offered such a unifying framework: We characterized the sense of control as the degree of *a priori* preparedness to act in future tasks, irrespective of the task itself. We formalized this characterization via a novel, interpretable measure of environmental controllability, Emp_ℓ , which quantifies how an environment’s structure supports flexible preparation for unknown future goals. This formulation unifies a broad range of existing constructs by showing that different perspectives on control reflect different measures of preparedness that an agent prospectively optimizes. The degree ℓ in Emp_ℓ characterizes this continuum: when $\ell = 1$, the measure reflects pure action availability (A-Control); values of $\ell > 1$ prioritize certainty in reaching outcomes (C_S-Control), while $\ell < 1$ emphasize preserving possibility (P_S-Control). Across three large-scale, theory-guided experiments, we showed that Emp_ℓ accurately predicts human preferences, offering a principled account of how environmental structure shapes the subjective sense of control. Together, these findings lay a foundation for a unified understanding of control, bridging objective and subjective perspectives and connecting insights across machine learning and psychology.

Empirically, we observed substantial individual differences in participants’ fitted values of ℓ . However, these differences showed no consistent associations with the trait-based measures LOC and IUS. This finding aligns with growing evidence that survey-based traits often lack associations with participants’ behavior in cognitive tasks^{66–68}. Yet, it raises the key question of what individual factors shape preferences for certainty versus possibility. In principle, individual differences along the certainty–possibility axis may relate to constructs from value-based decision-making, such as risk-seeking tendencies or exploratory behavior. However, our results suggest that this relationship is *a priori* non-trivial and requires empirical investigation. Specifically, Experiments 2 and 3 show that certainty–possibility preferences persist in punishment contexts as well as in explicit control judgments without instrumental framing. Moreover, this axis cannot be reduced to a conventional exploration–exploitation trade-off, as our paradigm lacks epistemic uncertainty or the need to learn action values. Hence, a promising direction for future work is to investigate how certainty–possibility preferences relate empirically to other psychological constructs, cognitive capacities, and personality traits.

Crucially, however, such investigations are not possible with existing paradigms which manipulate uniform action–outcome noise^{19,21,24,26} or use deterministic games^{22,69}. These approaches cannot disentangle certainty from possibility and thus cannot identify individual differences in ℓ (**Theorem 2** and **Proposition 3**). Our experimental paradigm is the first step for accessing this previously unmeasurable aspect of control sensitivity. Nevertheless, our paradigm has its own limitations. For example, by presenting participants with full knowledge of the environment, we intentionally removed the need for learning and exploration. Similarly, we performed the reward-seeking or punishment-avoiding tasks on behalf of participants, while they received feedback on our performance only once every seven trials; hence, we deliberately left their preferences unaffected by any notion of regret, which can dynamically influence preferences through counterfactual comparisons. Addressing these limitations remains a crucial next step toward a more comprehensive understanding of the human sense of control.

Moreover, our theoretical framework also has important implications for computer science and machine learning. In deep reinforcement learning (RL), the sense of control and empowerment has been used to learn controllable state representations^{70,71} and to support self-supervised skill discovery^{7,8,72}. States with high controllability have similarly been treated as intrinsically reward-

ing^{16,73–79}—linked to the curiosity-driven^{80–82} and intrinsically-motivated RL^{83–85}. However, the information-theoretic measures typically used in these settings^{7,8,16,70,72} lack transparent links to standard RL objectives, such as maximizing expected reward or minimizing the chance of failure. In light of recent efforts to clarify these connections¹⁰, we suggest that Emp_ℓ may serve as a novel and valuable building block for RL applications.

However, a limitation of our current formulation of Emp_ℓ is its focus on *immediate* preparedness and *one-step* reachability. In real-world settings, tasks typically require multiple intermediate actions, and any realistic notion of controllability must extend to multi-step horizons. Yet classical generalizations from control theory^{12,13} and information theory¹⁶ diverge in their multi-step formulations, and their underlying assumptions remain highly debated^{10,18,86}. A key direction for future research is thus to extend Emp_ℓ to multi-step settings—potentially in the context of goal-conditioned RL^{87–89}. A second limitation of Emp_ℓ is its assumption that the agent believes it knows the true dynamics of the environment. In most real-world scenarios, agents must contend with epistemic uncertainty, which must be accounted for in quantifying and pursuing control. Thus, another important direction is to adapt Emp_ℓ to settings with imperfect knowledge and to model the interplay between control-seeking and exploration (e.g., see ref.^{78,79}). Extending Emp_ℓ to such settings with imperfect knowledge would also allow the framework to capture the role of prior success and failure in shaping the subjective sense of control—including influencing the prior over a restricted set of states that the agents might be required to reach or avoid in the future. In realistic environments, agents form beliefs about reachability through experience rather than directly accessing the true dynamics. Phenomena such as learned helplessness can thus be understood as reduced inferred reachabilities, leading to a diminished perceived Emp_ℓ even when objective controllability is intact.

Along the same lines, we note that the human subjective sense of control depends on additional factors that go beyond the objective controllability that is defined based on action-outcome contingency^{6,31}. An important advantage of a unified and competent measure of the objective controllability of the environment is that it provides a basis for further formalization of these factors, thereby enabling a richer examination of the potentially dysfunctional psychological processes that might underlie biased subjective reports. This may aid our understanding of, for instance, how incorrect over-generalization may make us fall into helplessness even when we can objectively control the environment⁶⁰, or why we may experience illusions of control when none exists⁹⁰, if we perceive the mere availability of actions as generating controllability²⁷. Related to this, prior work has linked the mere availability of free choice to modulation of the dopaminergic mechanisms of RL²³; our framework provides a graded measure of controllability for future work to test how continuous variations in control modulate reward learning.

In conclusion, we introduced a formal and integrative framework for the human sense of control, grounded in the notion of *a priori* preparedness for unknown future tasks. By unifying divergent theoretical perspectives under a single model with just one interpretable parameter, our measure reconciles previously conflicting views on controllability and provides a principled link between objective environmental structure and subjective experience. Across three experiments, we showed that our measure not only captures key behavioral regularities but also reveals stable individual differences in how people value certainty versus possibility. These findings open up new avenues for understanding the foundations of agency, individual variation in control sensitivity, and their relevance across domains. Together, our results mark a step toward a unified science of control, one that can bridge minds and machines, theory and data, and certainty and possibility.

Methods

Experimental procedure

Ethics statement. All three experiments were approved by the ethics committee of the University of Tübingen (734/2019BO1). All participants consented to participation through an online consent form and a second form for the data protection protocol, presented at the beginning of the experiments.

Participants. All three experiments were implemented using the jsPsych toolkit⁹¹, and participants were recruited online from the Prolific platform ($n = 116$ for Experiment 1, $n = 307$ for Experiment 2, and $n = 200$ for Experiment 3, before exclusions); see below for justification of sample size. Only Experiment 3 was pre-registered (on December 10, 2025; <https://osf.io/rnf8v/>).

There were only three prescreening conditions: (i) fluency in English, (ii) age between 18 and 65, and (iii) a Prolific approval rate of higher than 95% (Experiment 1) or 99% (Experiments 2 and 3). For Experiment 2, we additionally excluded all participants who had participated in Experiment 1, and for Experiment 3, we additionally excluded all participants who had participated in Experiments 1 and 2. We did not impose any other pre-screening conditions, such as those based on participants’ nationality, country of residence, or educational background. Participants were paid a fixed participation fee plus a performance-contingent bonus that varied between the three experiments.

Experiment 1. Participants first practiced moving around in one deterministic and one probabilistic room (without any gold coin). Then, they played 10 rounds of the gold-collecting game in the same deterministic and probabilistic rooms. This was followed by an instruction about the room-selection procedure and how their choice of rooms would influence their bonus. Then, they practiced four room-selection trials followed by a comprehension test. Importantly, in one comprehension question, participants were presented with the *perfect* room, which had eight deterministic actions, where *all* peripheral locations were *certainly* reachable (i.e., equivalent to c_1 in the thought experiment). They were asked, ‘How could we increase the chance of gold-collecting in this room?’ and had three options: ‘By making actions probabilistic,’ ‘By adding more actions,’ and ‘The chance of collecting the gold is already maximum in this room.’ They could continue the experiment only after selecting the third option.

Participants performed 132 ($= 12 \times 11$) room-selection trials, in which they compared every pair of rooms twice (from the 12 predefined rooms; Supplementary Materials), with randomized spatial arrangements. The pairs were presented in shuffled order. Trials were aborted with a warning (and no bonus) if participants failed to choose a room within 10 seconds; no participant had more than 5 aborted trials. One of the 12 predefined rooms in the experiment was the perfect room mentioned above; hence, there were 22 trials in which participants had to choose between the perfect room and one of the other 11 rooms. We defined ‘perfect-room preference’ as the fraction of these 22 trials where the participants chose the perfect room. We excluded participants whose perfect-room preference was lower than the 95% quantile chance level (i.e., 0.7 for 22 trials). The remaining participants were included in our analyses ($n = 71$; 35 Female, 36 Male; age 34 ± 12). The excluded and included participants had clearly different performance statistics, separate from those used for exclusion (Supplementary Materials). This provides further confirmation that the exclusion criterion removed participants who indeed were not paying attention during the experiment.

Experiment 2. For the 156 participants who completed the gold block first, the general procedure for the gold block was identical to Experiment 1, with 9 rooms instead of 12 (Supplementary Materials). After completing the gold block, these participants were told that they would receive an additional 50 gold coins as a prize for their efforts and were asked to ‘protect their gold coins’ by avoiding the bombs. For participants who completed the bomb block first, the initial instructions were also similar to those in

Experiment 1, but bomb avoiding was the primary task. Initially, they were given 50 gold coins and were asked to protect the coins by avoiding the bombs. After completing the bomb block, they were told that they could collect more gold coins as a prize. For both conditions, and for both bomb and gold blocks, participants were required to complete a comprehension test. We included the question about the perfect room in all comprehension tests.

All participants completed a total of 144 ($=72 \times 2$) room selection trials. Trials were aborted with a warning (and no bonus) if participants failed to choose a room within 10 seconds; we excluded participants with more than 5 aborted trials in either of the gold or bomb blocks ($n = 4$). One of the 9 predefined rooms in the experiment was the perfect room mentioned above; hence, there were 16 trials in each block, where participants had to choose between the perfect room and one of the other 8 rooms. We excluded participants whose perfect-room preference was lower than the 95% quantile chance level (i.e., 0.75 for 16 trials) in either the gold or the bomb block. The remaining participants were included in our analyses ($n = 148$; 56 Female, 89 Male, 3 ‘Rather not say’; age 32 ± 10). The excluded and included participants had clearly different performance statistics, apart from those used for exclusion (Supplementary Materials). This provides further confirmation that the exclusion criterion removed participants who indeed were not paying attention during the experiment.

Experiment 3. Experiment 3 followed the general structure of Experiment 1, with two main differences. First, the room selection in Experiment 3 was framed as a judgment of controllability rather than a choice of room for gold collection. Second, the predefined set of rooms in Experiment 3 was identical to that used in Experiment 2 (9 rooms; Supplementary Materials), including the perfect room.

As in Experiment 1, participants first practiced moving around in one deterministic and one probabilistic room (without any gold coin). They then completed 10 rounds of the gold-collecting game in 10 different rooms. Unlike Experiment 1, performance in these rounds contributed to their bonus. The gold-collecting phase was introduced as the main task and was intended to provide participants with direct experience with the rooms’ transition structures—so that their subsequent judgments of controllability would be grounded in first-hand interaction. Participants were then explicitly informed that the bonus earned there was secured, and proceeded to a second phase. In this phase, participants compared pairs of rooms and indicated in which room they would feel more in control. They were told that there was generally no objectively correct answer and that we were interested in their subjective assessment. At the same time, to encourage attentive responding, participants were informed that in some cases the choice might be obvious. As an illustration, they were shown the perfect room and reminded: ‘In this room, you can go to any of the 8 locations that you may want, with 100% probability! So, you have as much control as you could possibly have!’ Participants were further informed that they would receive a one-off performance-based bonus for this phase, contingent on there being no indication of inattentive responding.

All participants completed 72 ($=9 \times 8$) control-judgment trials, corresponding to all pairwise comparisons among the 9 rooms. Trials were presented in randomized order and were aborted with a warning if no response was made within 10 seconds; no participant had more than 5 aborted trials. There were 16 trials in which participants had to choose between the perfect room and one of the other 8 rooms. We used the perfect-room preference across these trials both to determine bonus eligibility and as an exclusion criterion. Participants with a perfect-room preference greater than 50% received the one-off bonus, and those exceeding the 95% quantile chance level (i.e., 0.75 for 16 trials) were included in the analyses ($n = 174$; 86 Female, 87 Male, 1 ‘Rather not say’; age 40 ± 11). See Supplementary Materials for additional statistics.

Sample size estimation. The number of trials and the rooms were selected based on the model- and parameter-recovery analysis to dissociate different models and infer ℓ (Supplementary Materials). For Experiments 1 and 2, we targeted a sample size of at least 70 participants, yielding a maximum SEM of

approximately 5-6% for estimating the proportion of participants in a given group (i.e., $\sqrt{0.5^2/70} \approx 0.06$).

For Experiment 3, we conducted an additional power analysis based on the data from the 148 participants in Experiment 2 (pre-registered at <https://osf.io/rnf8v/>). For each candidate sample size $n \in [80, 140]$, we repeatedly (200 times) sub-sampled n participants with replacement, restricted the analysis to gain trials (matching the number of trials in Experiment 3), and repeated the Bayesian model selection shown in Fig. 3D1. Power was defined as the proportion of samples yielding a PXP greater than 0.95. This analysis indicated that 100 participants (after exclusions) were required to achieve a power of $> 80\%$. Accordingly, we recruited 200 participants. All details are provided in the pre-registration at <https://osf.io/rnf8v/>. However, subsequent convergence diagnostics indicated that the inference based on the Markov Chain Monte Carlo (MCMC) used in the preregistered power analysis had not converged, rendering the resulting sample-size estimate unreliable. We therefore repeated the resampling analysis using the corrected inference procedure described below and the data collected in Experiment 3. At the final sample size of $n = 174$, more than 99% of resampled datasets yielded a PXP greater than 0.95. Because this analysis was conducted retrospectively using the Experiment 3 data, we interpret this value as an empirical post hoc estimate of the experiment’s power. Equivalently, it quantifies the stability of the model-selection result under participant resampling.

Additional exclusion for surveys. Participants completed the LOC⁴¹ and IUS⁴² surveys at the end of all experiments. The LOC survey quantifies how much an individual attributes the causal sources of events to themselves (8 questions; I-LOC), to chance (8 questions; C-LOC), and to powerful others (8 questions; P-LOC), whereas IUS quantifies an individual’s degree of uncertainty intolerance and aversion (12 questions). In compliance with our ethics approval, the surveys did not influence participants’ bonuses, and the participants could skip any question they wished. Participants who skipped any questions were excluded from the analysis of the survey data. Additionally, we included three trivial questions (e.g., asking whether the participant could read) as attention checks. We also excluded participants who failed more than one attention check from further analysis (see ref.⁹²), as well as those exhibiting straight-line or zig-zag response patterns. For Experiments 1 and 2, this procedure left us a total of 177 (56 for Experiment 1 and 121 for Experiment 2) participants for the analysis of the survey data. For Experiment 3, it left us a total of 150.

Modeling and analyzing experimental data

General setting. We denote the participant’s response on trial t by $y_t \in \{1, 2\}$ and the pair of presented rooms by $x_t = (x_{t,1}, x_{t,2}) \in \{1, \dots, N_r\}^2$, where N_r is the total number of rooms (i.e., 12 for Experiment 1 and 9 for Experiment 2 and Experiment 3). We denote the data of participant n by $D_n = \{(x_t^{(n)}, y_t^{(n)})\}_{t=1}^{T_n}$, where T_n is the total number of trials that participant n completed. In the absence of aborted trials, $T_n = 132$ for Experiment 1 and $T_n = 72$ for each block of Experiment 2 as well as Experiment 3.

Model likelihood formulation. Given a model m and its parameters θ_m , the probability that a participant selects the first presented room (i.e., $y_t = 1$) is

$$p(y_t = 1 | x_t, \beta_m, \theta_m, m) = \frac{e^{\beta_m \cdot V_m(x_{t,1}; \theta_m)}}{e^{\beta_m \cdot V_m(x_{t,1}; \theta_m)} + e^{\beta_m \cdot V_m(x_{t,2}; \theta_m)}}, \quad (2)$$

where $V_m(x_{t,i}; \theta_m)$ is the ‘value’ assigned to room $x_{t,i}$ by model m , and $\beta_m \geq 0$ is the model inverse temperature that controls the sensitivity of decision-making to the model’s assigned values. For the random and N_a models, there is no parameter θ_m , and we have

$$V_{\text{Rand}}(x_{t,i}) = 0 \quad \text{and} \quad V_{N_a}(x_{t,i}) = N_a(x_{t,i}), \quad (3)$$

where $N_a(x_{t,i})$ denotes the number of actions available in room $x_{t,i}$. For the Emp_ℓ model, we have $\theta_{\text{Emp}} = \log \ell$ and

$$V_{\text{Emp}}(x_{t,i}; \ell) = \text{Emp}_\ell(x_{t,i}), \quad (4)$$

where $\text{Emp}_\ell(x_{t,i})$ is the Emp_ℓ value assigned to the central state in room $x_{t,i}$. For the general Bradley-Terry model^{43,44}, we have $\theta_{\text{Gen.}} \in \mathbb{R}^{N_r}$ and

$$V_{\text{Gen.}}(x_{t,i}; \theta_{\text{Gen.}}) = \theta_{\text{Gen.}, x_{t,i}}, \quad (5)$$

meaning that the value assigned to each room is considered as a free parameter⁴⁴. Without loss of generality, we always assumed $\theta_{\text{Gen.},1} = 0$, implying that $\theta_{\text{Gen.},j}$ quantifies the differential value of room j *relative* to room 1.

As is the typical assumption in pairwise comparisons^{44,53}, we assumed independence between different trials. This implies that the total log likelihood of D_n is

$$\log p(D_n | \beta_m, \theta_m, m) = \sum_{t=1}^{T_n} \log p(y_t^{(n)} | x_t^{(n)}, \beta_m, \theta_m, m). \quad (6)$$

Model selection. We first performed non-hierarchical inference, considering different participants to be fully independent of each other. Specifically, we considered that a participant's data D was generated according to the following generative model (see ref.^{49,93,94}):

$$\begin{aligned} m &\sim \text{Uniform}(\text{Rand}, N_a, \text{Emp}_\ell, \text{General}) \\ \beta_m | m &\sim \mathcal{N}_{\text{trunc}(0,\infty)}(0, 10) \\ \theta_m | m &\sim p_m \\ D_n | \beta_m, \theta_m, m &\sim p(\cdot | \beta_m, \theta_m, m), \end{aligned} \quad (7)$$

where $\mathcal{N}_{\text{trunc}(0,\infty)}(0, 10)$ is a truncated normal distribution (between 0 and ∞ , with mean zero and SD 10) acting as a rather uninformative prior for β_m ; the prior distribution p_m for the parameters θ_m will be presented separately for each model. The data likelihood $p(\cdot | \beta_m, \theta_m, m)$ is given by Eq. 6.

For the random and N_a models, there is no θ_m . For the Emp_ℓ model, we assumed $\theta_m = \log \ell$ with $p_m = \mathcal{N}(0, 2)$ as a rather broad prior. For the general model, we assumed that the components of $\theta_m \in \mathbb{R}^{N_r}$ are statistically independent, $\theta_{m,1} = 0$, and $\theta_{m,i} \sim \mathcal{N}(0, 1)$; the idea is that θ_m captures the scaled preferences between rooms (with a prior variance of 1), whereas β_m scales the preferences to drive action-selection.

For each participant n , we estimated the log model evidence $\log p(D_n | m_n)$ using both the Bayesian Information Criterion (BIC; a.k.a. Schwarz approximation⁹⁵) and bridge sampling⁹⁶. The resulting estimates showed strong agreement across all three experiments, with correlation coefficients ranging from 0.93 to 0.9997 (Supplementary Materials). Given this close agreement, we used the Schwarz approximation in the subsequent model-selection analyses. This choice makes our model-selection approach consistent with commonly used BIC-based methods⁴⁹ and reduces the potential sensitivity of the model-evidence estimates to the specification of parameter priors for the high-dimensional Bradley-Terry model. Model-recovery analyses⁴⁹ confirmed that the resulting model-selection procedure reliably recovered the true data-generating model across a range of parameters (see Supplementary Materials).

We then used the estimates of log-model evidence in the hierarchical generative model of ref.⁴⁷: Assuming that $m_n = m$ with some model frequency r_m that is shared across participants (see ref.^{47,49}). The frequencies were assumed to have a Dirichlet prior with uniform mean and 1/4 concentration parameters, with 4 being the number of models. Fig. 2E1, Fig. 3D1, and Fig. 5C1 show $\mathbb{E}[r_m | D_{1:N}]$ and

$\sqrt{\text{Var}[r_m|D_{1:N}]}$, where $D_{1:N}$ is the concatenated data of all N participants; we have $N = 71$ for Experiment 1, $N = 148$ for Experiment 2, and $N = 174$ for Experiment 3. PXP (Fig. 2E2, Fig. 3D1, and Fig. 5C2) denotes $\mathbb{P}[r_m > r_{m'} \forall m' \neq m | D_{1:N}]$, while additionally accounting for the null hypothesis that all models have the same frequencies⁴⁵. We assigned a winning model $\hat{m}_n$ to each participant, where we defined $\hat{m}_n = \arg \max_m \mathbb{P}(m_n = m | D_{1:N})$ (Supplementary Materials). Fig. 3D2 shows the fraction of the winning models. All inferences for the hierarchical model were performed through an MCMC procedure using the Metropolis-Hastings algorithm (an improved version of the one used in ref.⁹⁷).

Inferring ℓ . For inferring ℓ , we only included the Emp_ℓ and general participants, i.e., those with $\hat{m}_n \in \{\text{Emp}_\ell, \text{Gen.}\}$. For inferring the range of ℓ , we used the same 2-step model-selection procedure as above to infer a second model $m' \in \{\ell < 1, \ell = 1, \ell > 1\}$. For $\ell < 1$, we considered $\log \ell \sim \mathcal{N}_{\text{trunc}(-\infty, 0)}(0, 2)$, and, for $\ell > 1$, we considered $\log \ell \sim \mathcal{N}_{\text{trunc}(0, \infty)}(0, 2)$. Given that all three models were low-dimensional and differed only in their prior specification, we used the product-space method^{98,99} with MCMC sampling to evaluate the posterior model probabilities $p(m' | D_n)$. Inference was performed using a Gibbs sampling scheme that combined Hamiltonian Monte Carlo for $\log \ell$ and β with a Metropolis-Hastings update for the model variable m' , as implemented in `Turing.jl`¹⁰⁰; we tested the MCMC convergence based on the effective sample size and $\hat{r}$ ¹⁰¹, following the implementation in `Turing.jl`¹⁰⁰. Model recovery analysis⁴⁹ showed that the model selection based on these assumptions successfully recovered the true range of ℓ , even when the prior distributions for parameters did not match the exact ones we used (see Supplementary Materials). Fig. 2G1, Fig. 3G, and Fig. 5E1 show the expected model frequencies for this model selection.

Independently of the model selection for the range of ℓ , an additional inference was performed to evaluate $p(\ell_n | D_n)$ with the same prior as the one in the previous section, i.e., $\log \ell \sim \mathcal{N}(0, 2)$. The values in Fig. 2G2-G3, Fig. 4, and Fig. 5E2-E3 correspond to $\log \hat{\ell}_n := \mathbb{E}[\log \ell_n | D_n]$.

Statistical tests. All frequentist statistical tests were either a two-sided, one-sample paired t-test (Fig. 2F, Fig. 2H, Fig. 3E, Fig. 5D, and Fig. 5F) or a two-sided correlation test (Fig. 2G3, Fig. 4, Fig. 5E3). The correction for multiple hypotheses testing was done by controlling the FDR at 0.05⁴⁶ over all 45 null hypotheses that are presented in Fig. 2 (13), Fig. 3 (16), Fig. 4 (3), and Fig. 5 (13); p-value threshold: 0.05. All Bayes Factors (abbreviated BF in the figures) were evaluated using the Schwartz approximation¹⁰² to avoid any assumptions on the prior distribution. See Supplementary Materials for the detailed statistics of all tests.

Normalized accuracy rates

General theory. Given a participant, we denote the underlying data distribution by p_d , i.e., $p_d(y|x) \in [0, 1]$ represents the probability that a participant responds y when presented with the pair x of rooms. Similarly, a model of the participant is specified by p_m (as in Eq. 2). Given a pair x of rooms, the model predicts that the participant responds 1 if $p_m(1|x) > 0.5$. Otherwise, it predicts that the participant responds 0 if $p_m(0|x) > 0.5$. There will be a tie if $p_m(0|x) = 0.5$. Accordingly, the theoretical accuracy rate of the model is defined as^{50,103}

$$\begin{aligned} \alpha_m &:= \mathbb{E}[\mathbb{1}_{p_m(y|x) > 0.5}] \\ &= \underbrace{\sum_x p(x)}_{\text{averaged over all pairs}} \sum_y \underbrace{p_d(y|x)}_{\text{true probability of } y} \times \underbrace{\mathbb{1}_{p_m(y|x) > 0.5}}_{\text{whether the model predicts } y}. \end{aligned} \quad (8)$$

where $\mathbb{1}$ is the indicator function (with the abuse of notation $\mathbb{1}_{0.5 > 0.5} := 0.5$), and $p(x)$ is the experiment-dependent distribution of the pairs of presented rooms. We can similarly define the accuracy rate α_d of

the data distribution p_d as

$$\alpha_d := \mathbb{E}[\mathbb{1}_{p_d(y|x) > 0.5}] = \sum_x p(x) \sum_y p_d(y|x) \mathbb{1}_{p_d(y|x) > 0.5}. \quad (9)$$

It is straightforward to show that

$$\begin{aligned} \sum_y p_d(y|x) \mathbb{1}_{p_d(y|x) > 0.5} &= \max\{p_d(0|x), p_d(1|x)\} \\ &\geq \sum_y p_d(y|x) \mathbb{1}_{p_m(y|x) > 0.5} \end{aligned} \quad (10)$$

for any model m . Hence, we can recover a well-known result of statistical learning^{50,103} that $\alpha_m \leq \alpha_d$ for any model m . This implies that α_d is the maximum possible accuracy rate that we could achieve only if we knew the true data distribution p_d (or, at least, the choices of x where $p_d(1|x) > 0.5$). Accordingly, the ratio α_m/α_d quantifies the fraction of predictable actions that model m can predict correctly. The rest of this section aims to find a cross-validated estimate of α_m and an upper-bound estimate of α_d .

Cross-validated estimation of model accuracy rate. Given a model m , the empirical accuracy rate $\hat{\alpha}_m$ is often defined as^{50,103}

$$\hat{\alpha}_m := \frac{1}{T} \sum_{t=1}^T \mathbb{1}_{p_m(y_t|x_t) > 0.5}, \quad (11)$$

which is an unbiased estimation of the accuracy rate, i.e., $\mathbb{E}[\hat{\alpha}_m] = \alpha_m$. For each participant, model m can be any of our specified models that is *fitted* to the participant's data. We used two-fold cross-validation to find an unbiased estimate of α_m for such fitted models¹⁰⁴.

Since we have included each pair of rooms twice in the experiment, we can split the data D_n of participant n into two ‘independent’ replications. For the sake of notation, we define $N_{rr} = N_r(N_r - 1)/2$ to be the number of unique pairs of rooms and define $\{x_1, \dots, x_{N_{rr}}\}$ as the set of unique pairs of rooms. For example, we have $N_{rr} = 66$ for Experiment 1 and $N_{rr} = 36$ for each block of Experiment 2 as well as Experiment 3. Considering $y_{i,1}^{(n)}$ and $y_{i,2}^{(n)}$ to be the two responses of participant n to the presentation of the pair x_i , we split the data D_n into

$$D_{n,1} = \{(x_i, y_{i,1}^{(n)})\}_{i=1}^{N_{rr}} \quad \text{and} \quad D_{n,2} = \{(x_i, y_{i,2}^{(n)})\}_{i=1}^{N_{rr}}. \quad (12)$$

Then, for a given model m , we define

$$p_m(y|x) = \begin{cases} p(y|x, m, D_{n,2}), & \text{if } y \text{ belongs to } D_{n,1} \\ p(y|x, m, D_{n,1}), & \text{if } y \text{ belongs to } D_{n,2} \end{cases} \quad (13)$$

to be used in Eq. 11. To avoid any data contamination or assumption on ℓ , for the Emp_ℓ model, we used a broad prior $\log \ell \sim \mathcal{N}(0, 2)$ with no limitation of the model range. The inference was done using the Hamiltonian Monte Carlo algorithm as implemented in `Turing.jl`¹⁰⁰. The data points in the insets of Fig. 2G3, Fig. 4A2, and Fig. 5E3 show $\mathbb{E}[\log \ell_n | D_{n,2}]$ versus $\mathbb{E}[\log \ell_n | D_{n,1}]$.

Split half noise ceiling. We additionally used the splits in Eq. 12 to find a rough estimate of the maximum accuracy rate α_d as

$$\hat{\alpha}_d^+ := \frac{1}{2} \left(1 + \frac{1}{N_{rr}} \sum_i \mathbb{1}_{y_{i,1} = y_{i,2}} \right). \quad (14)$$

Assuming that $y_{i,1}$ and $y_{i,2}$ are statistically independent, it is straightforward to show that $\hat{\alpha}_d^+$ is an unbiased estimate of an upper bound on α_d :

$$0.5 \leq \alpha_d \leq \mathbb{E}[\hat{\alpha}_d^+] \leq \frac{1 + \alpha_d}{2} \leq 1. \quad (15)$$

The normalized accuracy rate in Fig. 2F, Fig. 3E, and Fig. 5D refers to $\hat{\alpha}_m / \hat{\alpha}_d^+$, where $\hat{\alpha}_m$ is evaluated by 2-fold cross validation. The noise ceiling of 1 corresponds to a model m that achieves $\hat{\alpha}_m = \hat{\alpha}_d^+$. The accuracy rates for $H_{\max}$ in Fig. 5D were not pre-registered.

Linking normative objectives to Emp_ℓ

General setting and notation. In this section, we show how normative objectives in Fig. 1C can be expressed in terms of Emp_ℓ . We consider an agent that interacts with an environment defined by a state space $\mathcal{S}$, a state-dependent action space $\mathcal{A}(s)$ at state s , and the probability $p(s'|s, a)$ for transition to state s' after taking action a in state s . We denote the number of states and actions by $N_s := |\mathcal{S}|$ and $N_a(s) := |\mathcal{A}(s)|$, respectively. Capital letters represent random variables; when no ambiguity arises, we omit the capital letter notation. The agent’s policy in state s is specified by the probability $\pi_s(a) := \pi(a|s)$ of selecting action a .

Preparedness for a to-be-announced goal state. To quantify the agent’s sense of control in a given state s , we start from the notion that the sense of control is the degree of ‘*a priori* preparedness for immediately collecting rewards or avoiding punishments,’ by being sensitive to any to-be-announced extrinsic reward function $R_{\text{ext}}(s)$. Specifically, we first consider the case in which $R_{\text{ext}}(s) = \mathbb{1}_{s=G}$ rewards agents upon visiting a goal state G , where G is *a priori* unknown—considered as a random variable with a uniform distribution over $\mathcal{S}$. This choice of reward closely resembles the typical setting in goal-conditioned RL^{87–89} but with the additional assumption that the ‘commanded goal’ G has a uniform distribution on $\mathcal{S}$; see ref.¹⁰⁵ for a conceptually similar idea.

Given this setting, the idea is thus to ask ‘Which state one should attempt to stay in to be best prepared to reach G in one step when the goal is announced?’ If the goal state G was known, then we could find the immediate optimal ‘value’ of state s :

$$\begin{aligned} V_*(s; G) &= \max_{a \in \mathcal{A}(s)} \sum_{s' \in \mathcal{S}} p(s'|s, a) R_{\text{ext}}(s') \\ &= \max_{a \in \mathcal{A}(s)} p(G|s, a) =: p_{\max}(G|s), \end{aligned} \quad (16)$$

where $p_{\max}(G|s)$ is the immediate reachability of state G from s . Hence, to be ‘ready’ to reach G when announced, the agent must locate itself in states with a high value of $V_*(s; G)$. However, since G is a random variable, $V_*(s; G)$ is also a random variable. Hence, being ‘ready’ to reach a goal state and, equivalently, the sense of control associated with s must be defined based on some statistics of $V_*(s; G)$.

One possibility is that the agent feels ‘ready’ when it is in a state with the highest *expected* value, where the expectation is taken over possible choices of the goal state G :

$$\mathbb{E}[V_*(s; G)] = \frac{1}{N_s} \sum_{s' \in \mathcal{S}} p_{\max}(s'|s) = \frac{\text{Emp}_1(s)}{N_s}, \quad (17)$$

where $\text{Emp}_1(s)$ is the degree-1 empowerment defined in Eq. 1. As is precisely stated in Proposition 1 below, this measure quantifies the number of effectively distinct actions; more intuitively, Corollary 1 shows that $\text{Emp}_1(s)$ can be seen as a regularized version of $N_a(s)$, where regularization accounts for overlap between

the outcome distributions of different actions. Specifically, if there are only two available actions, then $\text{Emp}_1(s)$ can be expressed as a well-known distance of $p(\cdot|s, a1)$ from $p(\cdot|s, a2)$, explicitly quantifying their distinction ([Corollary 2](#)). Hence, being ready in terms of expected future rewards resembles maximizing a notion of A-Control ([Fig. 1C](#)).

Proposition 1. *For $\ell = 1$, we have*

$$\text{Emp}_1(s) = 1 + \sum_{a \in \mathcal{A}(s)} \Delta(a; s), \quad (18)$$

where $\Delta(a; s)$ measures how distinct a is from the other actions:

$$\Delta(a; s) := \frac{1}{N_a(s)} \sum_{a' \in \mathcal{A}(s)} \left[\sum_{s' \in \mathcal{S}^*(a; s)} |p(s'|s, a) - p(s'|s, a')| \right] \quad (19)$$

with $\mathcal{S}^*(a; s) \subseteq \mathcal{S}$ the set of states where $a = \arg \max_{\tilde{a}} p(s'|s, \tilde{a})$; when there are ties, we assume that $\arg \max$ returns one action based on a given ordering.

Proof: See Supplementary Materials.

Corollary 1. *For $\ell = 1$ and $N_a(s) > 1$, we have*

$$\text{Emp}_1(s) = \underbrace{N_a(s)}_{\text{mere number of actions}} - \underbrace{\left(N_a(s) - 1 \right) \left(1 - \Delta(s) \right)}_{\text{regularization regarding action distinction}} \quad (20)$$

where $\Delta(s)$ measures the variability of actions in state s :

$$\Delta(s) := \frac{\sum_{a \in \mathcal{A}(s)} \Delta(a; s)}{N_a(s) - 1} \in [0, 1] \quad (21)$$

with $\Delta(a; s)$ defined in [Proposition 1](#). We have $\Delta(s) = 1$ if and only if $p(\cdot|s, a)$ have disjoint supports for different actions; this results in maximum degree-1 empowerment: $\text{Emp}_1(s) = N_a(s)$. We have $\Delta(s) = 0$ if and only if all actions have the exact same $p(\cdot|s, a)$; this results in minimum degree-1 empowerment: $\text{Emp}_1(s) = 1$.

Proof: See Supplementary Materials.

Corollary 2. *If $\mathcal{A}(s) = \{a1, a2\}$, then $\text{Emp}_1(s) = 1 + d_{\text{TV}}[p(\cdot|s, a1), p(\cdot|s, a2)]$, where d_{TV} denotes the total variation distance.*

Proof: See Supplementary Materials.

Another possibility is that the agent feels ‘ready’ when it is in a state where it has the *highest chance of certainly getting a reward*:

$$\mathbb{P}(V_*(s; G) = 1) = \frac{1}{N_s} \lim_{\ell \rightarrow \infty} \sum_{s' \in \mathcal{S}} p_{\max}^\ell(s'|s) = \frac{\text{Emp}_\infty(s)}{N_s}, \quad (22)$$

where $\text{Emp}_\infty(s) := \lim_{\ell \rightarrow \infty} \text{Emp}_\ell(s)$ is the degree- ∞ empowerment defined in [Eq. 1](#). This measure counts the total number of certainly reachable states from state s . Hence, being ready in terms of the highest chance of certain rewards resembles maximizing a notion of C_S-Control ([Fig. 1C](#)). A third possibility is

that the agent feels ‘ready’ if it is in a state where it has the *lowest chance of NOT getting a reward*:

$$\mathbb{P}(V_*(s; G) > 0) = \frac{1}{N_s} \lim_{\ell \rightarrow 0} \sum_{s' \in \mathcal{S}} p_{\max}^\ell(s'|s) = \frac{\text{Emp}_0(s)}{N_s}, \quad (23)$$

where $\text{Emp}_0(s) := \lim_{\ell \rightarrow 0} \text{Emp}_\ell(s)$ is the degree-0 empowerment defined in Eq. 1. This measure counts the total number of possibly (even if extremely unlikely) reachable states from state s ; hence, being ready in terms of the lowest chance of NOT getting rewards resembles maximizing a notion of P_S-Control (Fig. 1C).

Extension beyond a to-be-announced goal state. Eq. 17, Eq. 22, and Eq. 23 show how Emp_ℓ relates to different normative objectives, under the assumption that there is only a single rewarding state. Here we show that the correspondence of Emp_ℓ to different normative objectives holds for a much broader class of reward distributions. Specifically, we consider a distribution over state-based reward functions $R_{\text{ext}} : \mathcal{S} \rightarrow \mathbb{R}$ with the following constraints (see Supplementary Materials for the precise statement): 1. The distribution over reward function R_{ext} is invariant to permutations of R_{ext} over states (i.e., there is no *a priori* preference for some particular states). 2. There is almost surely only one best state. **Theorem 1** below shows that increasing Emp_ℓ increases both upper and lower bounds of different normative objectives.

Theorem 1. *Suppose the distribution of the reward function R_{ext} admits the assumptions stated above. We denote the maximum reward by the random variable*

$$R_{\text{ext}}^* := \max_{s \in \mathcal{S}} R_{\text{ext}}(s)$$

and the most negative reward value by the random variable

$$R_{\text{ext}}^{-*} := \min_{s \in \mathcal{S}} R_{\text{ext}}(s) \mathbb{1}_{\min_{s \in \mathcal{S}} R_{\text{ext}}(s) < 0}.$$

Then, there exist constants c_0, C_0, c_1, C_1 , and $C_\infty \in \mathbb{R}^+$ and b_1 and $B_1 \in \mathbb{R}$, dependent solely on the distribution of the reward function, such that the following holds.

1. *The probability of a certain maximum reward is proportional to degree- ∞ empowerment:*

$$\mathbb{P}(V_*(s; R_{\text{ext}}) = R_{\text{ext}}^* > 0) = C_\infty \text{Emp}_\infty(s). \quad (24)$$

2. *Degree-1 empowerment bounds the expected reward:*

$$b_1 + c_1 \text{Emp}_1(s) \leq \mathbb{E}[V_*(s; R_{\text{ext}})] \leq B_1 + C_1 \text{Emp}_1(s) \quad (25)$$

3. *Degree-0 empowerment bounds the probability of possibly receiving a reward higher than the minimum negative reward:*

$$c_0 \text{Emp}_0(s) \leq \mathbb{P}(V_*(s; R_{\text{ext}}) > R_{\text{ext}}^{-*}) \leq C_0 \text{Emp}_0(s) \quad (26)$$

For all bounds, equality holds if $R_{\text{ext}}(s) = \mathbb{1}_{s=G}$ with $G \sim \text{Unif}(\mathcal{S})$.

Proof: See Supplementary Materials.

Remark 1. *All results of **Theorem 1** hold if the realized rewards are stochastic, and $R_{\text{ext}} : \mathcal{S} \rightarrow \mathbb{R}$ denotes their state-based expectation.*

Corollary 3. *If $R_{\text{ext}}(s) = \mathbb{1}_{s=G} - 1$ with $G \sim \text{Unif}(\mathcal{S})$, then we have $\mathbb{E}[V_*(s; R_{\text{ext}})] = \text{Emp}_1(s)/N_s - 1$ for the expected reward, $\mathbb{P}(V_*(s; R_{\text{ext}}) = 0) = \text{Emp}_\infty(s)/N_s$ for the probability of a guaranteed no-punishment, and $\mathbb{P}(V_*(s; R_{\text{ext}}) > -1) = \text{Emp}_0(s)/N_s$ for the probability of possibly no-punishment.*

Proof: See Supplementary Materials.

Emp_ℓ as a function of self-inefficacy and noisy action execution

The notion of self-efficacy^{29,31} is primarily concerned with the extent of control an agent feels over executing its desired actions. It is important to note that control in the sense of ‘achieving desired outcomes’ is distinct from control in the sense of ‘executing desired actions’, although these are related^{29,31}. Here, we show that introducing self-inefficacy in terms of action-execution noise naturally decreases Emp_ℓ. This demonstrates that quantifying controllability via Emp_ℓ is conceptually consistent with these psychological theories^{29,31}.

Formally, we distinguish between an agent’s intended action a in state s and the executed action a_{exc} . We can then model noisy and erroneous action execution by assuming that a_{exc} is most of the time (with probability $1 - \epsilon$) equal to the intended action a , but is sometimes (in the case of error, with probability ϵ) sampled from a noise distribution π_{noise} independently of the agent’s intended action, i.e.,

$$\pi_{\text{exc}}(a_{\text{exc}}|s, a) := (1 - \epsilon) \mathbb{1}_{a_{\text{exc}}=a} + \epsilon \pi_{\text{noise}}(a_{\text{exc}}|s). \quad (27)$$

Accordingly, if the environment transition probabilities, given the executed action a_{exc} , are described by p , then the agent’s perceived transition probabilities, given its intended actions, are described by

$$p_\epsilon(s'|s, a) = (1 - \epsilon) p(s'|s, a) + \epsilon p_{\text{noise}}(s'|s), \quad (28)$$

where $p_{\text{noise}}(s'|s) := \sum_{a'} p(s'|s, a') \pi_{\text{noise}}(a'|s)$ is the next state distribution in the case of a noisy and erroneous action selection.

This formulation connects the execution of desired actions to the achievability of desired outcomes. Specifically, our formalism closely matches the original conceptualization of self-efficacy in psychology³¹, and the equivalents of our noise probability ϵ have indeed been used previously as operationalizations of self-inefficacy in experimental studies^{21,106}. **Theorem 2** below shows that, for all ℓ , the perceived degree- ℓ empowerment of state s decreases by increasing the probability ϵ of the execution noise. **Theorem 2** additionally shows that Emp_ℓ accounts for the causation-prediction distinction proposed by ref.¹⁷ as a necessary property of formal measures of the sense of control.

Theorem 2. *Suppose the action execution is noisy as in Eq. 27. Then, the perceived degree- ℓ empowerment of state s , derived based on $p_\epsilon(s'|s, a)$ in Eq. 28 and denoted by $\text{Emp}_{\ell, \epsilon}(s)$, is a decreasing function of ϵ , i.e.,*

$$\frac{\partial \text{Emp}_{\ell, \epsilon}(s)}{\partial \epsilon} \leq 0 \quad (29)$$

for all $s \in \mathcal{S}$, $\epsilon \in [0, 1)$, $\ell > 0$, π_{noise} , and p .

Proof: See Supplementary Materials.

Existing measures of the sense of control

In this section, we review different existing measures of the sense of control in Fig. 1C. We also demonstrate how the various measures relate to different conceptualizations.

Mere number of actions. The simplest existing measure is the mere number $N_a(s)$ of actions available in state s (e.g., see ref.¹). This serves as a crude measure of A-Control. In the toy example of Fig. 1A, an agent’s sense of control based on $N_a(s)$ follows the ranking in Fig. 1B2. However, the main issue with $N_a(s)$ is that it does not account for the degree to which the actions are distinct. For example, duplicating the action to the painting class in c_2 would increase its ‘controllability’ according to $N_a(s)$. Two more appropriate alternatives are the (1-step) Klyubin-empowerment¹⁶ $\text{Emp}_{\text{Kly}}(s)$ and the maximum outcome entropy¹⁴ $H_{\text{max}}(s)$; we discuss these alternatives next.

Klyubin-empowerment. The (1-step) Klyubin-empowerment $\text{Emp}_{\text{Kly}}(s)$ is defined as the maximum mutual information between the next state S' and the current action A , conditioned on the current state $S = s$:

$$\begin{aligned} \text{Emp}_{\text{Kly}}(s) &:= \max_{\pi_s} \underbrace{I_{\pi}(S', A|S = s)}_{\text{transfer entropy}^{17}} \\ &= \max_{\pi_s} \left(\underbrace{H_{\pi}[A|S = s]}_{\text{action diversity}} - \underbrace{H_{\pi}[A|S', S = s]}_{\text{regularization regarding action distinction}} \right), \end{aligned} \quad (30)$$

where I_{π} and H_{π} denote mutual information and entropy, respectively, under policy π ; we note that, in the 1-step case considered here, Eq. 30 is equivalent to other formulations and generalizations of the Klyubin-empowerment^{18,86}. In the reformulation of mutual information (the second equality in Eq. 30), the first term $H_{\pi}[A|S = s]$ quantifies the action diversity at state s and is a proxy of the total number of actions at state s ; e.g., its maximum value is $\log N_a(s)$. On the other hand, $H_{\pi}[A|S', S = s]$ quantifies the ambiguity of action A in light of S and S' (i.e., how accurately the agent can infer its past action based on the transition). Hence, $e^{\text{Emp}_{\text{Kly}}(s)} \in [1, N_a(s)]$ can be seen as a regularized version of $N_a(s)$ and a measure of A-Control, similarly to $\text{Emp}_1(s)$ (Corollary 1). ref.¹⁰ presents an alternative formulation of this intuition through the lens of information geometry.

In the toy example of Fig. 1A, an agent’s sense of control based on $\text{Emp}_{\text{Kly}}(s)$ follows the ranking in Fig. 1B2. The mutual information $I_{\pi}(S', A|S = s)$ is also known as (conditional) transfer entropy, which is another proposed measure of the sense of control¹⁷. It is straightforward to show that, for any choice of π , an agent’s sense of control based on transfer entropy also follows the ranking in Fig. 1B2. Hence, conceptually, Emp_{Kly} (and transfer entropy) quantifies a notion of action-distinction and hence A-Control.

However, action distinction is the only quality of a state that Emp_{Kly} quantifies. Specifically, Emp_{Kly} does not consider the outcome distribution of different actions *as long as they are distinct*. Proposition 2 below makes this statement precise, connects Emp_{Kly} to the Emp_1 , and showcases that Emp_{Kly} is blind to certainty-possibility dichotomy.

Proposition 2. Suppose $\mathcal{A}(s)$ consists of $M(s)$ partitions $\{\mathcal{A}_1(s), \dots, \mathcal{A}_{M(s)}(s)\}$, where: 1. All actions within each partition are identical, i.e., $p(\cdot|s, a) = p(\cdot|s, a')$ if a and a' are in the same partition. 2. Actions in different partitions lead to distinct sets of next states, i.e., $p(\cdot|s, a)$ and $p(\cdot|s, a')$ have disjoint supports on $\mathcal{S}$ if a and a' belong to different partitions. Then, we have $\text{Emp}_1(s) = e^{\text{Emp}_{\text{Kly}}(s)} = M(s)$, independently of the exact transition probabilities.

Proof: See Supplementary Materials.

Maximum outcome entropy. The (1-step) maximum entropy $H_{\text{max}}(s)$ quantifies the diversity of the potential next states that an agent can reach:

$$H_{\text{max}}(s) := \max_{\pi_s} H_{\pi}[S'|S = s], \quad (31)$$

where $H_\pi[S'|S = s]$ denotes the Shannon entropy of S' under policy π (Fig. 1C). In the toy example of Fig. 1A, an agent’s sense of control based on $H_{\max}(s)$ follows the ranking in Fig. 1B3. Accordingly, $H_{\max}(s)$ can be seen as a measure of C_S-Control, but it is important to note that $H_{\max}(s)$ ignores any notion of certainty of outcomes or distinctiveness of actions.

Sensing control in deterministic states

A deterministic state is uniquely characterized by the transition function $T(., s) : \mathcal{A}(s) \rightarrow \mathcal{S}$ that determines the next state s' given the previous state-action pair s, a , i.e.,

$$p(s'|s, a) = \mathbb{1}_{s'=T(s,a)}. \quad (32)$$

We call two deterministic actions ‘distinct’ regarding their outcome if they take the agent to different next states, i.e., if $T(s, a) \neq T(s, a')$; otherwise, the two actions are indistinguishable regarding their outcome. As a result, in a deterministic state, the number of distinct actions is equal to the number of reachable next states expressed as $|\{T(s, a) : a \in \mathcal{A}(s)\}|$. Proposition 3 below shows that, in deterministic states, different measures of the sense of control are indistinguishable from each other. Notably, they all assign the highest value of the sense of control to central hubs^{16,22,79}.

Proposition 3. *Suppose $s \in \mathcal{S}$ is a ‘deterministic’ state with transition function T as in Eq. 32. Then, we have*

$$\text{Emp}_\ell(s) = e^{\text{Emp}_{\text{KlY}}(s)} = e^{H_{\max}(s)} = |\{T(s, a) : a \in \mathcal{A}(s)\}| \quad (33)$$

for all $\ell \geq 0$.

Proof: See Supplementary Materials.

Acknowledgement

We thank Sophia Becker, Rahul Bhui, Marcel Binz, Franziska Brändle, Johanni Brea, Wulfram Gerstner, Lucas Gruaz, Milena Rmus, Kai Sandbrink, Valentin Schmutz, and Elena Zamfir for insightful discussions on related topics and/or helpful comments on the manuscript. We also thank Susanne Haridi, Theresa Horn, Tianyuan Teng, Andrew Webb, Kristin Witte, and many other members of the Dayan Lab and the Institute for Human-Centered AI for their help with the experiment design and data collection. This research was supported by the Max Planck Society (PD), the Humboldt Foundation (PD), and the European Research Council (ES).

Data availability

All data needed to reproduce the results reported in this manuscript are publicly available at <https://github.com/modirshanechi/SenseOfControl>. The pre-registration of Experiment 3 is available at <https://osf.io/rnf8v/>.

Code availability

All code needed to reproduce the results reported in this manuscript is publicly available at <https://github.com/modirshanechi/SenseOfControl>.

Author Contributions Statement

All authors developed the study concept. AM did the formal analysis, conducted the experiments, and collected and analyzed the data. All authors wrote the paper.

Competing Interests Statement

The authors declare no competing interests.

References

1. Leotti, L. A., Iyengar, S. S. & Ochsner, K. N. Born to choose: The origins and value of the need for control. *Trends in Cognitive Sciences* **14**, 457–463 (2010).
2. Moscarello, J. M. & Hartley, C. A. Agency and the calibration of motivated behavior. *Trends in Cognitive Sciences* **21**, 725–735 (2017).
3. Wang, K. S., Yang, Y.-Y. & Delgado, M. R. How perception of control shapes decision making. *Current Opinion in Behavioral Sciences* **41**, 85–91 (2021).
4. Goddu, M. K. & Gopnik, A. The development of human causal learning and reasoning. *Nature Reviews Psychology* **3**, 319–339 (2024).
5. Nussenbaum, K. & Hartley, C. A. Understanding the development of reward learning through the lens of meta-learning. *Nature Reviews Psychology* **3**, 424–438 (2024).
6. Ly, V., Wang, K. S., Bhanji, J. & Delgado, M. R. A reward-based framework of perceived control. *Frontiers in Neuroscience* **13** (2019).
7. Eysenbach, B., Gupta, A., Ibarz, J. & Levine, S. Diversity is all you need: Learning skills without a reward function. In *International Conference on Learning Representations* (2019).
8. Choi, J., Sharma, A., Lee, H., Levine, S. & Gu, S. S. Variational empowerment as representation learning for goal-conditioned reinforcement learning. In *International Conference on Machine Learning*, 1953–1963 (PMLR, 2021).
9. Leibfried, F., Pascual-Díaz, S. & Grau-Moya, J. A unified Bellman optimality principle combining reward maximization and empowerment. In Wallach, H. *et al.* (eds.) *Advances in Neural Information Processing Systems*, vol. 32 (Curran Associates, Inc., 2019).
10. Myers, V., Ellis, E., Levine, S., Eysenbach, B. & Dragan, A. Learning to assist humans without inferring rewards. In *Advances in Neural Information Processing Systems* (2024).
11. Du, Y. *et al.* AvE: Assistance via Empowerment. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M. & Lin, H. (eds.) *Advances in Neural Information Processing Systems*, vol. 33, 4560–4571 (Curran Associates, Inc., 2020).
12. Sontag, E. D. *Mathematical Control Theory: Deterministic Finite Dimensional Systems* (Springer New York, NY, 2013).
13. Ogata, K. *Modern Control Engineering* (Prentice Hall, 2010), 5th edn.
14. Huys, Q. J. & Dayan, P. A Bayesian formulation of behavioral control. *Cognition* **113**, 314–328 (2009).
15. Cover, T. M. *Elements of Information Theory* (John Wiley & Sons, 1999).

16. Klyubin, A., Polani, D. & Nehaniv, C. Empowerment: a universal agent-centric measure of control. In *2005 IEEE Congress on Evolutionary Computation*, vol. 1, 128–135 Vol.1 (2005).
17. Ligneul, R. Prediction or causation? Towards a redefinition of task controllability. *Trends in Cognitive Sciences* **25**, 431–433 (2021).
18. Abel, D. *et al.* Plasticity as the mirror of empowerment. *arXiv preprint arXiv:2505.10361* (2025).
19. Mancinelli, F., Roiser, J. & Dayan, P. Internality and the internalisation of failure: Evidence from a novel task. *PLoS Computational Biology* **17**, e1009134 (2021).
20. Dorfman, H. M. & Gershman, S. J. Controllability governs the balance between pavlovian and instrumental action selection. *Nature Communications* **10**, 5826 (2019).
21. Sandbrink, K., Hunt, L. & Summerfield, C. Understanding human meta-control and its pathologies using deep neural networks. *PsyArXiv* (2024).
22. Brändle, F., Stocks, L. J., Tenenbaum, J. B., Gershman, S. J. & Schulz, E. Empowerment contributes to exploration behaviour in a creative video game. *Nature Human Behaviour* (2023).
23. Cockburn, J., Collins, A. G. & Frank, M. J. A reinforcement learning mechanism responsible for the valuation of free choice. *Neuron* **83**, 551–557 (2014).
24. Grahek, I., Frömer, R., Prater Fahey, M. & Shenhav, A. Learning when effort matters: neural dynamics underlying updating and adaptation to changes in performance efficacy. *Cerebral Cortex* **33**, 2395–2411 (2022).
25. Ligneul, R., Mainen, Z. F., Ly, V. & Cools, R. Stress-sensitive inference of task controllability. *Nature Human Behaviour* **6**, 812–822 (2022).
26. Frömer, R., Lin, H., Dean Wolf, C., Inzlicht, M. & Shenhav, A. Expectations of reward and efficacy guide cognitive control allocation. *Nature Communications* **12**, 1030 (2021).
27. Stolz, D. S., Müller-Pinzler, L., Krach, S. & Paulus, F. M. Internal control beliefs shape positive affect and associated neural dynamics during outcome valuation. *Nature Communications* **11**, 1230 (2020).
28. Limbachia, C. *et al.* Controllability over stressor decreases responses in key threat-related brain areas. *Communications Biology* **4**, 42 (2021).
29. Ajzen, I. Perceived behavioral control, self-efficacy, locus of control, and the theory of planned behavior. *Journal of Applied Social Psychology* **32**, 665–683 (2002).
30. Seligman, M. E. & Maier, S. F. Failure to escape traumatic shock. *Journal of Experimental Psychology* **74**, 1–9 (1967).
31. Bandura, A. Self-efficacy: toward a unifying theory of behavioral change. *Psychological Review* **84**, 191–215 (1977).
32. Rotter, J. B. Generalized expectancies for internal versus external control of reinforcement. *Psychological Monographs: General and Applied* **80**, 1–28 (1966).
33. Kool, W., McGuire, J. T., Rosen, Z. B. & Botvinick, M. M. Decision making and the avoidance of cognitive demand. *Journal of Experimental Psychology: General* **139**, 665 (2010).
34. Shenhav, A., Botvinick, M. M. & Cohen, J. D. The expected value of control: an integrative theory of anterior cingulate cortex function. *Neuron* **79**, 217–240 (2013).

35. Iyengar, S. S. & Lepper, M. R. When choice is demotivating: Can one desire too much of a good thing? *Journal of Personality and Social Psychology* **79**, 995 (2000).
36. Chernev, A., Böckenholt, U. & Goodman, J. Choice overload: A conceptual review and meta-analysis. *Journal of Consumer Psychology* **25**, 333–358 (2015).
37. Sutton, R. S. & Barto, A. G. *Reinforcement Learning: An Introduction* (MIT press, 2018).
38. Puterman, M. L. *Markov Decision Processes: Discrete Stochastic Dynamic Programming* (John Wiley & Sons, 1994).
39. Bertsekas, D. *Dynamic Programming and Optimal Control: Volume I* (Athena Scientific, 2012).
40. Twitter, Inc. Twemoji: Twitter emoji (2025). URL <https://github.com/twitter/twemoji>.
41. Levenson, H. Activism and powerful others: Distinctions within the concept of internal-external control. *Journal of Personality Assessment* **38**, 377–383 (1974).
42. Carleton, R. N., Norton, M. P. J. & Asmundson, G. J. Fearing the unknown: A short version of the intolerance of uncertainty scale. *Journal of Anxiety Disorders* **21**, 105–117 (2007).
43. Bradley, R. A. & Terry, M. E. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika* **39**, 324–345 (1952).
44. Agresti, A. *Categorical Data Analysis* (John Wiley & Sons, 2013).
45. Rigoux, L., Stephan, K. E., Friston, K. J. & Daunizeau, J. Bayesian model selection for group studies—revisited. *NeuroImage* **84**, 971–985 (2014).
46. Efron, B. & Hastie, T. *Computer age statistical inference* (Cambridge University Press, 2016).
47. Stephan, K. E., Penny, W. D., Daunizeau, J., Moran, R. J. & Friston, K. J. Bayesian model selection for group studies. *NeuroImage* **46**, 1004–1017 (2009).
48. MacKay, D. J. *Information theory, inference and learning algorithms* (Cambridge university press, 2003).
49. Wilson, R. C. & Collins, A. G. Ten simple rules for the computational modeling of behavioral data. *eLife* **8**, e49547 (2019).
50. Hastie, T., Tibshirani, R., Friedman, J. H. & Friedman, J. H. *The elements of statistical learning: data mining, inference, and prediction*, vol. 2 (Springer, 2009).
51. Kahneman, D. & Tversky, A. Prospect theory: An analysis of decision under risk. *Econometrica* **47**, 263–291 (1979).
52. Pachur, T., Schulte-Mecklenbeck, M., Murphy, R. O. & Hertwig, R. Prospect theory reflects selective allocation of attention. *Journal of Experimental Psychology: General* **147**, 147–169 (2018).
53. Peterson, J. C., Bourgin, D. D., Agrawal, M., Reichman, D. & Griffiths, T. L. Using large-scale experiments and machine learning to discover theories of human decision-making. *Science* **372**, 1209–1214 (2021).
54. Alloy, L. B. & Abramson, L. Y. Judgment of contingency in depressed and nondepressed students: Sadder but wiser? *Journal of Experimental Psychology: General* **108**, 441–485 (1979).
55. Linser, K. & Goschke, T. Unconscious modulation of the conscious experience of voluntary control. *Cognition* **104**, 459–475 (2007).

56. Sato, A. Both motor prediction and conceptual congruency between preview and action-effect contribute to explicit judgment of agency. *Cognition* **110**, 74–83 (2009).
57. Chambon, V. & Haggard, P. Sense of control depends on fluency of action selection, not motor performance. *Cognition* **125**, 441–451 (2012).
58. Constant, M., Salomon, R. & Filevich, E. Judgments of agency are affected by sensory noise without recruiting metacognitive processing. *Elife* **11**, e72356 (2022).
59. Wenke, D., Fleming, S. M. & Haggard, P. Subliminal priming of actions influences sense of control over effects of action. *Cognition* **115**, 26–38 (2010).
60. Baratta, M. V., Seligman, M. E. & Maier, S. F. From helplessness to controllability: toward a neuroscience of resilience. *Frontiers in Psychiatry* **14**, 1170417 (2023).
61. Maier, S. F. & Seligman, M. E. Learned helplessness at fifty: Insights from neuroscience. *Psychological Review* **123**, 349–367 (2016).
62. Beck, B., Di Costa, S. & Haggard, P. Having control over the external world increases the implicit sense of agency. *Cognition* **162**, 54–60 (2017).
63. Giersiepen, M., Schütz-Bosbach, S. & Kaiser, J. Freedom of choice boosts midfrontal theta power during affective feedback processing of goal-directed actions. *Biological Psychology* **183**, 108659 (2023).
64. Deci, E. L. & Ryan, R. M. The support of autonomy and the control of behavior. *Journal of Personality and Social Psychology* **53**, 1024 (1987).
65. Gregor, K., Rezende, D. J. & Wierstra, D. Variational intrinsic control. *arXiv preprint arXiv:1611.07507* (2016).
66. Anvari, F., Billinger, S., Analytis, P. P., Franco, V. R. & Marchiori, D. Testing the convergent validity, domain generality, and temporal stability of selected measures of people’s tendency to explore. *Nature Communications* **15**, 7721 (2024).
67. Witte, K., Thalmann, M. & Schulz, E. How should we measure exploration? *PsyArXiv* (2024).
68. Vrizzi, S., Najar, A., Lemogne, C., Palminteri, S. & Lebreton, M. Behavioral, computational and self-reported measures of reward and punishment sensitivity as predictors of mental health characteristics. *Nature Mental Health* (2025).
69. Lidayan, A. *et al.* Intrinsically-motivated humans and agents in open-world exploration. *arXiv preprint arXiv:2503.23631* (2025).
70. Bharadhwaj, H., Babaeizadeh, M., Erhan, D. & Levine, S. Information prioritization through empowerment in visual model-based RL. In *International Conference on Learning Representations* (2022).
71. Pathak, D., Agrawal, P., Efros, A. A. & Darrell, T. Curiosity-driven exploration by self-supervised prediction. In *Proceedings of the 34th International Conference on Machine Learning - Volume 70*, ICML’17, 2778–2787 (JMLR.org, 2017).
72. Shelhamer, E., Mahmoudieh, P., Argus, M. & Darrell, T. Loss is its own reward: Self-supervision for reinforcement learning. *arXiv preprint arXiv:1612.07307* (2016).
73. Salge, C., Glackin, C. & Polani, D. Empowerment—an introduction. *Guided Self-Organization: Inception* 67–114 (2014).

74. Volpi, N. C. & Polani, D. Goal-directed empowerment: combining intrinsic motivation and task-oriented behavior. *IEEE Transactions on Cognitive and Developmental Systems* **15**, 361–372 (2020).
75. Becker-Ehmck, P., Karl, M., Peters, J. & van der Smagt, P. Exploration via empowerment gain: Combining novelty, surprise and learning progress. In *ICML 2021 Workshop on Unsupervised Reinforcement Learning* (2021).
76. Cao, H. *et al.* Towards empowerment gain through causal structure learning in model-based reinforcement learning. In *The Thirteenth International Conference on Learning Representations* (2025).
77. Jung, T., Polani, D. & Stone, P. Empowerment for continuous agent—environment systems. *Adaptive Behavior* **19**, 16–39 (2011).
78. Dai, S., Xu, W., Hofmann, A. & Williams, B. An empowerment-based solution to robotic manipulation tasks with sparse rewards. *Autonomous Robots* **47**, 617–633 (2023).
79. Gruaz, L., Modirshanechi, A., Becker, S. & Brea, J. Merits of curiosity: a simulation study. *PsyArXiv (to appear in Open Mind)* (2024).
80. Modirshanechi, A., Kondrakiewicz, K., Gerstner, W. & Haesler, S. Curiosity-driven exploration: foundations in neuroscience and computational modeling. *Trends in Neurosciences* **46**, 1054–1066 (2023).
81. Gottlieb, J. & Oudeyer, P.-Y. Towards a neuroscience of active sampling and curiosity. *Nature Reviews Neuroscience* **19**, 758–770 (2018).
82. Oudeyer, P.-Y. Computational theories of curiosity-driven learning. *arXiv preprint arXiv:1802.10546* (2018).
83. Singh, S., Lewis, R. L., Barto, A. G. & Sorg, J. Intrinsically motivated reinforcement learning: An evolutionary perspective. *IEEE Transactions on Autonomous Mental Development* **2**, 70–82 (2010).
84. Aubret, A., Matignon, L. & Hassas, S. A survey on intrinsic motivation in reinforcement learning. *arXiv preprint arXiv:1908.06976* (2019).
85. Oudeyer, P.-Y. & Kaplan, F. What is intrinsic motivation? a typology of computational approaches. *Frontiers in Neurobotics* **1**, 6 (2009).
86. Capdepuy, P. *Informational principles of perception-action loops and collective behaviours*. Ph.D. thesis, University of Hertfordshire (2011).
87. Eysenbach, B., Salakhutdinov, R. & Levine, S. C-learning: Learning to achieve goals via recursive classification. In *International Conference on Learning Representations* (2021).
88. Eysenbach, B., Zhang, T., Levine, S. & Salakhutdinov, R. Contrastive learning as goal-conditioned reinforcement learning. In Oh, A. H., Agarwal, A., Belgrave, D. & Cho, K. (eds.) *Advances in Neural Information Processing Systems* (2022).
89. Mendonca, R., Rybkin, O., Daniilidis, K., Hafner, D. & Pathak, D. Discovering and achieving goals via world models. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P. & Vaughan, J. W. (eds.) *Advances in Neural Information Processing Systems*, vol. 34, 24379–24391 (Curran Associates, Inc., 2021).
90. Langer, E. J. The illusion of control. *Journal of Personality and Social Psychology* **32**, 311 (1975).
91. De Leeuw, J. R., Gilbert, R. A. & Luchterhandt, B. jspsych: Enabling an open-source collaborative ecosystem of behavioral experiments. *Journal of Open Source Software* **8**, 5351 (2023).

92. Zorowitz, S., Solis, J., Niv, Y. & Bennett, D. Inattentive responding can induce spurious associations between task behaviour and symptom measures. *Nature Human Behaviour* **7**, 1667–1681 (2023).
93. Daw, N. Trial-by-trial data analysis using computational models. *Decision making, affect, and learning: Attention and performance XXIII* **23** (2011).
94. Barber, D. *Bayesian Reasoning and Machine Learning* (Cambridge University Press, 2012).
95. Schwarz, G. Estimating the dimension of a model. *The Annals of Statistics* **6**, 461–464 (1978).
96. Gronau, Q. F. *et al.* A tutorial on bridge sampling. *Journal of mathematical psychology* **81**, 80–97 (2017).
97. Modirshanechi, A., Lin, W.-H., Xu, H. A., Herzog, M. H. & Gerstner, W. Novelty as a drive of human exploration in complex stochastic environments. *Proceedings of the National Academy of Sciences* **122**, e2502193122 (2025).
98. Carlin, B. P. & Chib, S. Bayesian model choice via markov chain monte carlo methods. *Journal Of The Royal Statistical Society Series B: Statistical Methodology* **57**, 473–484 (1995).
99. Lodewyckx, T. *et al.* A tutorial on bayes factor estimation with the product space method. *Journal of Mathematical Psychology* **55**, 331–347 (2011).
100. Ge, H., Xu, K. & Ghahramani, Z. Turing: a language for flexible probabilistic inference. In *International Conference on Artificial Intelligence and Statistics, AISTATS 2018, 9-11 April 2018, Playa Blanca, Lanzarote, Canary Islands, Spain*, 1682–1690 (2018).
101. Vehtari, A., Gelman, A., Simpson, D., Carpenter, B. & Burkner, P.-C. Rank-normalization, folding, and localization: An improved r-hat for assessing convergence of mcmc. *Bayesian Analysis* **16**, 667–718 (2021).
102. Kass, R. E. & Raftery, A. E. Bayes factors. *Journal of the American Statistical Association* **90**, 773–795 (1995).
103. Bishop, C. M. *Pattern Recognition and Machine Learning* (Springer New York, NY, 2006).
104. Bates, S., Hastie, T. & Tibshirani, R. Cross-validation: What does it estimate and how well does it do it? *Journal of the American Statistical Association* **119**, 1434–1445 (2024).
105. Pezzulo, G. & Ognibene, D. Proactive action preparation: Seeing action preparation as a continuous and proactive process. *Motor Control* **16**, 386–424 (2012).
106. Sandbrink, K. & Summerfield, C. Modelling cognitive flexibility with deep neural networks. *Current Opinion in Behavioral Sciences* **57**, 101361 (2024).